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AgriWatch: A Unified Satellite-Based Agricultural Early-Warning System

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Abstract

Desert locust invasions and in-season crop stress threaten Pakistan's food security, yet both are normally detected only after damage is visible. This report presents AgriWatch, a satellite-based screening and early-warning prototype with two modules built from freely available Earth-observation data. The locust module has built a labelled dataset for the Cholistan–Tharparkar desert margin from FAO observation records (2008–2026) and satellite vegetation and climate series (2001–2020), and will train a convolutional LSTM to forecast locust presence four to eight weeks ahead, evaluated by rolling-origin forecasts over the 2019–2020 upsurge. Preliminary baselines show that vegetation and weather anomalies alone do not beat chance on this data, whereas a rule based on recent sightings does; the ConvLSTM will be judged against that stronger baseline. The crop module applies a stage-aware Isolation Forest to Sentinel-2 wheat time series to rank fields by abnormal development. This report specifies requirements, methodology and design for both modules, reports an audit of the FAO data and the baseline results of the locust module; the ConvLSTM and the crop module have not yet been evaluated.

Executive Summary

Agriculture in Pakistan is exposed to two hazards that are normally studied separately: sudden pest invasions, the most destructive of which is the desert locust, and slower in-season stress in standing crops such as wheat. Both tend to be recognised late—after a swarm has been sighted or a field has visibly discoloured—when the cheapest control options are gone. FAO's published damage scenarios for the 2020 locust situation in Pakistan put potential losses at PKR 205 billion (about USD 1.3 billion) if 15% of wheat, gram and potato production were lost, and at USD 2.2 billion (rabi) and USD 2.9 billion (kharif) at a 25% damage level [1, 2]. In a four-year surveillance of the Pak-Afghan bordering Hindukush region, wheat yellow rust was found in 83% of surveyed fields, with 29% showing severity of 50% or more [3].

Vision. AgriWatch is a research prototype that asks one question twice: *can free satellite and climate data show that a place is behaving abnormally early enough to be useful?* The two modules answer it at two scales, and both are built only from data that can be downloaded from a computer—no field equipment, drones or hyperspectral cameras.

Locust Risk Module. Historical FAO observation records are the labels; satellite vegetation, soil moisture, temperature and rainfall anomalies are the inputs. The study region is the Cholistan–Tharparkar desert margin (24–31°N, 68–74°E), divided into 0.25° cells (28×24 cells) at a 16-day time step. A small convolutional LSTM (about 2.7×10^5 parameters) will forecast locust presence at any life stage (breeding-stage presence is a secondary target) 28, 42 and 56 days ahead. In this module the region is selected by coordinates alone, so the box also covers part of Indian Rajasthan; satellite and climate data run from 2001 to 2020, anomalies are measured against a 2001–2019 climatology, models are trained on forecasts whose targets fall in 2008–2019, and the 2019–2020 upsurge is used for evaluation by rolling-origin forecasts in which every model is refitted only on labels already available at forecast time. Baselines of increasing complexity, including a rule based on recent sightings, are described in Section 5.6.4.

Crop Abnormality Screening Module. Wheat-field locations come from the public ADB ground-truth survey; Sentinel-2 Level-2A imagery provides the spectral time series. Each field is described by robust temporal features of ten spectral bands and five vegetation and moisture indices (15 variables × 8 temporal summaries = 120 features before filtering) and compared only with agronomically similar peers (same district, season and crop stage). In this module the survey covers a single season (Rabi 2022–23) and the data are partitioned by district, never by season. An Isolation Forest returns a continuous anomaly score. A Normal/Abnormal decision is claimed only after calibration against independently

verified Pakistani field observations, and the module is explicitly a screening alert, not a disease diagnosis.

Key design decisions. (i) Hyperspectral leaf-level imaging was dropped in favour of Sentinel-2 because the team has neither the cameras nor field access. (ii) Labels are self-constructed where tractable (locust) and not required where impossible to obtain from home (crop training). (iii) A deep contrastive crop model considered earlier was replaced by an Isolation Forest, which has a published parcel-level precedent [4] and is reproducible on a laptop. (iv) Claims are tied to evidence tiers, and negative results will be reported.

Work completed in FYP-I. A literature review of 25 studies and datasets; verification of every data source; a first audit of the FAO observation files, which found 887 Pakistani records inside the study box and several data-quality traps (Section 5.5.3); a software requirements specification (21 functional requirements, seven use cases, three dashboard screens and a data model), a complete methodology with a leakage-controlled evaluation protocol, a system design and a risk register. For the locust module the multi-year FAO records were then retrieved through the open catalogue, the dataset (inputs, masked labels, splits) was built and tested, and the baselines were evaluated by rolling-origin forecasts (Section 5.6.7). The ConvLSTM, the per-cell LSTM and the whole crop module have not yet been trained, so no performance claim is made for them.

Sustainable Development Goals. The project targets SDG 2 (Zero Hunger), SDG 13 (Climate Action) and SDG 1 (No Poverty).

Main risks and FYP-II plan. The two largest risks are (a) the scarcity and non-stationarity of locust records before the 2019–2020 upsurge (about 1,300 any-stage and only 63 hopper or band cell-steps in 2008–2018), which is why the environment-only baselines do not yet beat chance, and (b) access to locally verified crop-health labels. Each has a documented contingency (Section 4.10). FYP-II (about 16 weeks) covers the ConvLSTM and per-cell LSTM, the negative-example ablations, crop feature extraction and Isolation Forest scoring, held-out evaluation, a dashboard prototype and the final report (Section 5.9.4).

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Chapter 1 Introduction

Pakistan's agriculture faces two kinds of early-warning problem that are rarely tackled together. The first is a *sudden, regional* hazard: the desert locust (*Schistocerca gregaria*), whose swarms formed and invaded Balochistan, Sindh and Punjab during the 2019–2020 upsurge and prompted the Government of Pakistan to declare a national emergency in early 2020 [5]. FAO reported that breeding was under way across 38% of Pakistan's land area (60% in Balochistan, 25% in Sindh and 15% in Punjab) and warned that the whole country could be invaded if breeding was not contained [2]. The second is a *slow, local* hazard: in-season stress in standing crops such as wheat, caused by disease, water deficit, nutrient imbalance or late sowing, which lowers yield without being obvious until symptoms are visible.

In both cases the response is usually reactive. Locust control is organised once swarms or hopper bands are found, even though FAO's own forecasts reach only about six weeks ahead and rely on national survey teams reporting what they have seen [6]. Crop inspection depends on field visits that are expensive to repeat across large areas, and a four-year surveillance in the Pak-Afghan bordering Hindukush region found yellow rust in 83% of surveyed fields [3]. Free Earth-observation archives now make it possible to look for earlier signals: anomalous soil moisture and vegetation flush before locust breeding, and atypical spectral development in a wheat field before it is visibly damaged.

This project, **AgriWatch**, builds a research prototype around that idea. It has two modules that share a philosophy—*detect a departure from expected behaviour from satellite and climate data*—but not an architecture. The **Locust Risk Module** is a supervised spatiotemporal forecaster trained on a dataset we construct from FAO observation records and satellite and climate series. The **Crop Abnormality Screening Module** is an unsupervised anomaly-scoring pipeline that ranks wheat fields by how differently they develop from comparable fields. Everything is derived from data that can be downloaded from a laptop, which was a deliberate design decision (Section 5.3).

1.1 Purpose of this Document

The purpose of this document is to present the problem formulation, literature review, proposed methodology, system design and preliminary implementation of AgriWatch, our Final Year Project at FAST-NUCES Lahore. It is the Deliverable II report of this R&D project and follows the six prescribed chapters: introduction, project vision, literature review and related applications, software requirement specifications, proposed approach and methodology, and high-level and low-level design. It covers *both* modules of AgriWatch—the Locust Risk Module and the Crop Abnormality Screening Module—as one project. It records what has been decided and verified so far and fixes the evaluation protocol that the

next phase will execute. Measurements are reported only for the locust module: the audit of the FAO observation data (Section 5.5.3) and the results of its baseline models (Section 5.6.7). No ConvLSTM or crop-module result is reported.

The report is organised around four research questions:

RQ1. Can a convolutional LSTM, trained on a gridded dataset constructed from FAO records and satellite/climate anomalies, forecast locust presence (any life stage; breeding stage as a secondary target) 28–56 days ahead in Cholistan and Tharparkar better than a climatological prior, logistic regression, a tree ensemble, a per-cell LSTM and a rule based on recent sightings, when evaluated by rolling-origin forecasts over the 2019–2020 invasion?

RQ2. How sensitive are the locust results to the way negative examples are produced (survey-confirmed absences versus pseudo-absences)?

RQ3. Can a stage-aware Isolation Forest on Sentinel-2 features rank Pakistani wheat fields by abnormal development better than a peer-median deviation score and an NDVI-only rule?

RQ4. How much locally verified evidence is needed before a Normal/Abnormal decision for wheat fields can be claimed?

1.2 Intended Audience

The primary audience is the FYP supervisor and evaluation panel, who will judge whether the problem is well posed, the design is sound and the evaluation is honest. The second audience is future FAST students who may continue the work: the report is written so that the data pipeline, splits and metrics can be reproduced from it. The third is domain stakeholders—locust-control and plant-protection staff, provincial agriculture departments and agronomists—who may care about the outputs but not about the models; the Project Vision chapter is written for them.

1.3 Definitions, Acronyms, and Abbreviations

ADB	Asian Development Bank (publisher of the ground-truth crop survey)
DLIS	Desert Locust Information Service (FAO)
EO	Earth Observation
FAO	Food and Agriculture Organization of the United Nations
FYP	Final Year Project (FYP-I: first semester; FYP-II: second semester)

GEE	Google Earth Engine
Rabi / Kharif	Winter (about Oct–May) and summer (about May–Oct) cropping seasons in Pakistan
SDG	Sustainable Development Goal
UET	University of Engineering and Technology (Peshawar)
Upsurge	A period of widespread locust breeding and swarm formation across several countries

Technical terms (indices, model names, metrics) are defined in Section 3.1.

1.4 Conclusion

This chapter introduced the two hazards AgriWatch addresses, the common idea behind the two modules, the purpose and audience of the report, and four research questions. The remainder of the report is organised as follows. Chapter 2 sets out the project vision. Chapter 3 reviews related research and related applications. Chapter 4 specifies the software requirements of the whole system, including use cases, the dashboard interface, the data model and the risk analysis. Chapter 5 specifies the proposed approach and methodology of both modules, including data, labels, models, evaluation, a preliminary data audit and the work plan. Chapter 6 gives the high-level and low-level design.

Chapter 2 Project Vision

Locust swarms and crop stress look like different problems, but both are *detection-before-damage* problems that can be approached with the same raw material: freely available satellite and climate archives. This chapter states what AgriWatch is for, the problem it addresses, its objectives and boundaries, the Sustainable Development Goals it supports, the constraints it works under, and who it serves.

2.1 Problem Domain Overview

AgriWatch operates in the domain of *agricultural early warning from Earth observation*. Two hazards are considered.

Desert locust. The desert locust breeds in the arid margins of North Africa, the Middle East and Southwest Asia. After rain, moist sandy soils and green vegetation allow egg-laying; eggs hatch after about two weeks and hoppers need about six weeks to fledge into adults that can fly in swarms [6]. National survey teams report sightings to FAO, which combines them with weather, habitat and satellite information to issue monthly bulletins and forecasts up to six weeks ahead [6]. Breeding conditions depend on soil moisture, vegetation and temperature [7, 8], all of which satellites observe.

Wheat stress. Wheat is Pakistan's principal winter (rabi) crop. Disease, water deficit, nutrient deficiency, waterlogging, heat and late sowing all change canopy chlorophyll, density and water content, which changes how the field reflects light. Sentinel-2 observes visible, red-edge, near-infrared and shortwave-infrared bands at 10–20 m every few days, so a field's spectral trajectory over a season can be compared with that of its neighbours [9, 4].

2.2 Problem Statement

Locust. Pakistan's locust early warning depends on ground surveys and bulletin-based forecasts, and no open, Pakistan-specific, machine-learning risk model validated on Pakistan's own 2019–2020 invasion was found in the literature reviewed. Existing learned models are trained and tested on African breeding grounds [10] or on a multi-country tabular dataset [11].

Wheat. Pakistan has credible crop-type surveys and wheat-disease surveillance, but the public resources are not packaged as a dataset linking field location, date, a verified Normal/Abnormal health label and the matching Sentinel-2 time series. A supervised disease classifier trained on foreign data would not solve this, because crop time series shift between regions with variety, climate, sowing date and management [12].

Unifying problem. How can free satellite and climate data be used to flag abnormal conditions in Pakistan early, at regional scale for locusts and at field scale for wheat, without field equipment and with claims that match the available evidence?

2.3 Problem Elaboration

The problem decomposes into sub-problems, each of which has a counterpart in the methodology.

P1. Presence-only labels. FAO records say where locusts *were seen*; they do not say where they were absent. A cell without a record is *unknown*, not negative. Naive sampling of negatives teaches a model to separate desert from non-desert rather than breeding from non-breeding.

P2. Spatiotemporal forecasting. Breeding risk depends on conditions in neighbouring cells and on how they evolved over months; a model must use both space and time and predict a fixed number of weeks ahead.

P3. Leakage in evaluation. Adjacent 16-day steps and neighbouring cells are nearly identical, so random splits overstate performance; baselines fitted on test years would be unfair.

P4. Scarcity and non-stationarity of positives (locust module). Locust activity is episodic. Between 2008 and 2018 the study box holds about 1,300 positive cell-steps at any life stage (against about 19,700 cell-steps with a survey-confirmed absence) and only 63 for hoppers or hopper bands, almost all in 2010; the 2019–2020 upsurge is far richer. This limits what a deep model can learn and means that the training years do not resemble the evaluation years.

P5. Normal phenology versus abnormality. Wheat fields sown at different times or in different districts legitimately differ; an anomaly detector must not flag normal variation.

P6. No local health labels. Public Pakistani wheat data identify crop type, location and stage but not health. A spectral anomaly alone is not a disease label; disease, drought, nutrient stress, waterlogging and senescence overlap spectrally [13].

P7. Smallholder fields and mixed pixels. Fields can be close to the size of a few 10 m pixels, so boundary pixels contaminate field statistics [14].

P8. Integration. The two outputs must be presented together for a defined region without implying that they are the same kind of evidence.

2.4 Goals and Objectives

Goal. To build and rigorously evaluate a satellite-only early-warning prototype that flags abnormal locust-breeding conditions and abnormal wheat development in Pakistan, and to be explicit about what each result does and does not show.

Objectives.

O1. Construct a gridded locust dataset (inputs and masked labels) for the Cholistan–Tharparkar region (a coordinate box that also covers part of Indian Rajasthan) from FAO records and satellite and climate series, and document it so it can be released.

O2. Implement a baseline ladder (climatological prior, logistic regression, tree ensemble, per-cell LSTM, recent-sightings rule) and a ConvLSTM forecaster with three horizons (28, 42, 56 days).

O3. Evaluate the locust models on the 2019–2020 invasion by rolling-origin forecasts, with time-blocked development splits and a per-horizon embargo, and compare three ways of producing negative examples.

O4. Build a Pakistani wheat cohort from public crop-ground-truth data and Sentinel-2 Level-2A imagery, with documented inclusion and exclusion rules.

O5. Extract robust temporal features and fit a stage-aware Isolation Forest that produces a continuous field-level anomaly score, compared with a peer-median deviation score and an NDVI-only rule.

O6. Calibrate a Normal/Abnormal threshold on independently verified Pakistani observations when they can be obtained, and otherwise report anomaly ranking with Precision@k on inspected fields, never claiming accuracy that was not measured.

O7. Present both outputs in a dashboard prototype for a defined study region and, where the data allow, compare how early each module's signal appears.

2.5 Project Scope

In scope.

- A research prototype of two modules, with code, configuration and documentation sufficient to reproduce the experiments.
- The locust module for the Cholistan–Tharparkar region (24–31°N, 68–74°E) at 0.25° and 16-day resolution, with presence targets (any life stage; breeding stage as a secondary target) at 28, 42

and 56 days ahead.

- The crop module for wheat observed in the Pakistani ADB ground-truth survey (Rabi 2022–23), using Sentinel-2 Level-2A only.
- A leakage-controlled evaluation protocol applied identically to all models and baselines.
- A dashboard prototype showing locust risk maps and ranked field alerts.

Deliverables. The constructed locust dataset and its documentation; trained models and baselines with results (FYP-II); the wheat cohort manifest, feature definitions and anomaly scores; the dashboard prototype; this report and the FYP-II report.

Out of scope.

- New field, drone or hyperspectral data collection by the team, and any ground-sensor hardware.
- Disease identification, lesion segmentation, severity regression and treatment or pesticide recommendations.
- Real-time operation: the locust module is validated against historical events, and the dashboard is not a public service.
- Weather-forecast fusion and Sentinel-1 radar for the crop module.
- Claims of operational deployment readiness.

FYP-I versus FYP-II. FYP-I (this report) covers literature review, requirements, source verification, design, the data audit, the locust dataset and its baselines, and the evaluation protocol. FYP-II covers the ConvLSTM and the remaining locust models, the crop module's data extraction and scoring, final evaluation, the dashboard and the final analysis.

2.6 Sustainable Development Goal (SDG)

AgriWatch targets three Sustainable Development Goals; the full list is shown in Figure 2.1.

2.6.1 SDG 2: Zero Hunger

The primary goal. Target 2.4 calls for resilient agricultural practices that help maintain ecosystems and strengthen capacity to adapt to extreme weather and other disasters. Locust swarms and crop stress both threaten the production of staple crops; earlier and better-targeted warning supports earlier survey, control and inspection, and can reduce the area that has to be treated or visited. The wheat module



Figure 2.1: The 17 Sustainable Development Goals. *Description for screen readers:* the image is a grid of 17 coloured square tiles in three rows, each with a number, an icon and a goal name, followed by the words "Sustainable Development Goals". Row 1: 1 No Poverty (red), 2 Zero Hunger (mustard yellow, a bowl icon), 3 Good Health and Well-being (green, a heartbeat line), 4 Quality Education (dark red, an open book), 5 Gender Equality (red-orange), 6 Clean Water and Sanitation (light blue). Row 2: 7 Affordable and Clean Energy (yellow), 8 Decent Work and Economic Growth (maroon), 9 Industry, Innovation and Infrastructure (orange), 10 Reduced Inequalities (magenta), 11 Sustainable Cities and Communities (amber), 12 Responsible Consumption and Production (gold-brown). Row 3: 13 Climate Action (dark green, an eye over a globe), 14 Life Below Water (blue), 15 Life on Land (light green), 16 Peace, Justice and Strong Institutions (navy blue, a dove), 17 Partnerships for the Goals (dark blue, interlinked circles). AgriWatch targets goals 1, 2 and 13.

contributes to monitoring the crop that dominates the rabi season.

2.6.2 SDG 13: Climate Action

Target 13.1 concerns resilience and adaptive capacity to climate-related hazards. Locust breeding is driven by rainfall, soil moisture and temperature anomalies, and desert locust upsurges are linked to unusual weather in the breeding areas; drought and heat also shape wheat stress [15, 16]. A system that reads climate anomalies from open data to flag agricultural hazards is a climate-adaptation tool.

2.6.3 SDG 1: No Poverty

Target 1.5 concerns the resilience of the poor to climate-related extreme events and other shocks. Smallholder households bear the largest share of crop losses when a hazard is detected late; the UN estimated that about 34,000 households would need emergency livelihood and food-security assistance because of crop losses [5]. Earlier warning is a means to reduce such shocks.

2.7 Constraints

- **Data access.** The multi-year FAO archive is retrieved through an open catalogue whose date range must still be unlocked, and a formal access request to the FAO Locust Hub was submitted

in September 2026 and was pending at the time of writing. Locally verified wheat-health labels are request-only.

- **No fieldwork.** The team has no hyperspectral camera, drone or field access, so every input must be downloadable and no label can be created by field inspection.
- **Compute.** Training is limited to a laptop and free-tier GPUs; large foundation-model fine-tuning is out of reach in the core plan.
- **Time.** The project spans two semesters and is carried out by three students; FYP-I contains no model training.
- **Label scarcity and bias.** FAO records reflect survey effort as well as locust presence; wheat crop surveys are single-season and label crop type, not health.
- **Resolution.** A Sentinel-2 pixel (10 m) is about the size of a few square metres of canopy, not a leaf or lesion, and smallholder fields are small.
- **Licensing.** The FAO data were requested under terms that include research or academic use only, acknowledgement of FAO and sharing of publications; the project is non-commercial.

2.8 Business Opportunity

AgriWatch is a research project and is not built as a product, but its components have clear potential users. Locust-control and plant-protection agencies could use a gridded risk map to prioritise surveys in a country where survey area is large and resources are limited. Provincial agriculture departments and extension services could use field rankings to decide which fields to inspect first. Crop insurers and agricultural lenders need credible, scalable field-condition signals, which currently depend on sparse visits. Any commercial reuse would require its own review of data licences, because the FAO locust archive was requested for research and academic use and the crop-health labels would be obtained under the data holders' permissions. The most realistic path from this project is therefore a public, documented dataset and open pipeline, followed by pilots with institutional partners.

2.9 Stakeholders Description/ User Characteristics

2.9.1 Stakeholders Summary

Table 2.1: Stakeholders and their roles

Stakeholder	Role	What they need from AgriWatch
FAO / Desert Locust Information Service	Publishes observation data, bulletins and forecasts	Credit and data-use compliance; an independently validated regional risk model
Plant-protection and locust-control agencies (Pakistan)	Survey and control operations	A risk map that says where to survey first and how far ahead
Provincial agriculture departments and extension staff	Crop monitoring and advice	A ranked list of fields worth inspecting, with the reason (anomalous index, stage)
Farmers and farmer groups	Ultimate beneficiaries	Timely, understandable alerts through intermediaries
Agronomists and plant pathologists	Verify field condition	Anomaly ranking that reduces random inspection; clear limits on what an alert means
Insurers and agricultural lenders	Condition monitoring at scale	Transparent, auditable field signals
Researchers and future FYP teams	Reuse and extend	Documented dataset, code and evaluation protocol
FYP supervisor and panel	Evaluation	A well-posed problem, sound design and honest evidence

2.9.2 Key High-Level Goals and Problems of Stakeholders

Operational users want *earlier* and *more targeted* action: knowing a few weeks ahead which grid cells are likely to host breeding, and which wheat fields are developing unusually, lets scarce survey and inspection capacity be used first where it matters. Their main problem is trust: a map or ranking that cannot be explained, or whose accuracy was never measured on Pakistani events, will not be acted on. This is why the project fixes its evaluation on a real held-out Pakistani invasion, reports precision-recall rather than accuracy under class imbalance, and states alerts as inspection priorities rather than diagnoses.

2.10 Conclusion

AgriWatch addresses two detection-before-damage problems with a shared philosophy and honest evaluation. The scope is a research prototype, restricted to what can be built and validated from open data by three students in two semesters, and aligned with SDG 2, SDG 13 and SDG 1. The next chapter reviews the literature that motivates the design.

Chapter 3 Literature Review and Related Applications

This chapter reviews 25 studies and datasets and four related applications that bear on the design of AgriWatch. The studies are grouped by the question each answers: how locust breeding has been predicted from satellite data; how spatiotemporal deep learning handles gridded sequences; how crop anomalies have been detected without labels; how wheat stress and disease have been monitored with Sentinel-2; what Pakistani data and evidence exist; and how models transfer across regions. Each study is summarised, critically assessed and related to the proposed work. The applications are reviewed in Section 3.4. Reported results come from different sensors, regions, labels and splits and are not directly comparable; they are quoted to show what each study established, not to rank them.

3.1 Definitions, Acronyms, and Abbreviations

Anomaly (z-score) A value expressed in standard deviations from its usual (climatological) value at the same place and time of year.

Breeding-stage record An FAO record of hoppers or hopper bands, which cannot fly and therefore indicate local hatching.

Climatology The long-term mean and spread of a variable for each place and time of year.

ConvLSTM Convolutional Long Short-Term Memory network: an LSTM whose states are spatial maps updated by convolutions [17].

Embargo A gap removed between training and evaluation periods so that no training example's target overlaps the evaluation period.

ERA5-Land A global land reanalysis (daily temperature, precipitation and soil moisture at about 0.1°) used for three locust-module channels [18].

GNDVI / NDRE / NDMI / NDVI / OSAVI Spectral indices defined in Section 5.7.4.

Isolation Forest (IF) An unsupervised anomaly detector that isolates points by random splits; points isolated in fewer splits are more anomalous [19].

L2A Sentinel-2 Level-2A: atmospherically corrected surface reflectance.

MODIS, MOD13Q1 Moderate Resolution Imaging Spectroradiometer; the MOD13Q1 product gives 16-day, 250 m vegetation indices [20].

NIR / SWIR Near-infrared and shortwave-infrared parts of the spectrum.

Peer group Fields of the same district (or agro-ecological zone), season and crop stage, used as the reference for "normal".

PR-AUC Area under the precision–recall curve; preferred to ROC-AUC when positives are rare [21].

Precision@k Fraction of the k highest-scoring fields that are confirmed abnormal.

Persistence baseline A forecast with no fitted parameters that scores each cell by how recently and how often locusts were recorded in it and its neighbours.

Presence-only data Data that record where an event was observed but never where it was confirmed absent.

Pseudo-absence An artificially chosen location treated as a negative example in presence-only modelling [22].

Red edge The steep rise in leaf reflectance between red and NIR wavelengths; sensitive to chlorophyll.

Rolling-origin evaluation Forecasts are issued at successive origins through the evaluation period; each is scored once, by a model fitted only on labels that were already known at that origin.

SCL Sentinel-2 Scene Classification Layer (cloud, shadow, snow, vegetation, etc. per pixel).

3.2 Detailed Literature Review

3.2.1 Desert Locust Prediction from Satellite and Climate Data

3.2.1.1 Klein et al. and Kimathi et al. — environmental drivers of breeding [23, 8]

Summary. Klein et al. review how satellite data have been applied to locust research and management, covering the environmental variables that characterise habitat and the operational uses of such data. Kimathi et al. fit a MaxEnt model on desert locust presence records (9,134 occurrences) from other parts of the species' range, using temperature, rainfall, sand and moisture contents as predictors, and project it onto East Africa; ten-day greenness maps from February to April 2020 are overlaid to show how breeding locations evolved.

Critical analysis. The review maps what has been done rather than testing a method, and it predates deep sequence and foundation models. The MaxEnt suitability map is essentially static: time enters

only through an overlay, so it learns neither temporal dynamics nor a lead time. Together they make the choice of input variables defensible.

Relation to this work. They support using vegetation, soil moisture, temperature and rainfall as the inputs of the locust module (Section 5.6.1), and Kimathi et al. show the risk of transferring models across regions, which is why we train on Pakistani data rather than reuse an African model.

3.2.1.2 Ellenburg et al. — soil moisture and texture thresholds [7]

Summary. Using FAO observations from the East Africa upsurge (October 2019–June 2020), the authors combine soil texture and model-derived soil moisture into "optimal thresholds" of breeding conditions. When all thresholds were met, the conditions matched the estimated egg-laying period 62.5% of the time, rising to 85.2% when a 3×3 pixel buffer accounted for data uncertainty; including soil moisture cut the area of optimal conditions from 33% to under 20% on average.

Critical analysis. The approach is interpretable and physically grounded, and it shows that soil moisture adds real discriminating power. It is a set of thresholds fitted to one region and one upsurge, not a learned forecaster, and it gives no explicit forecast horizon. The study area is East Africa, not Pakistan.

Relation to this work. Soil moisture is a core input channel. The buffered score motivates reporting a spatially tolerant metric (Section 5.6.6), since a cell adjacent to a record is a near-hit, not a miss.

3.2.1.3 Khan et al. — LocustLens [11]

Summary. FAO locust attack records are fused with climate variables into a 42-country dataset (GLAD42), and machine-learning classifiers are used for swarm prediction.

Critical analysis. Wide geographic coverage and a public dataset are real strengths. The data are tabular, so spatial neighbourhoods are not preserved and a model that reads a map of the region cannot be used; fine lead-time prediction is not its aim.

Relation to this work. We reproduce this style of model as a baseline (tree ensemble, Section 5.6.4) on our own grid, rather than using GLAD42 as model input.

3.2.1.4 Yusuf et al. — geospatial prediction of breeding grounds [10]

Summary. The authors curate a dataset from FAO observation records and use two kinds of spatiotemporal input: remotely sensed environmental and climate data, and multispectral Earth-observation images. They fine-tune the Prithvi geospatial foundation model and report an F1 score of 81.53% for breeding-ground prediction in Africa.

Critical analysis. It is the closest deep-learning precedent and is operationally minded, but its result is for African breeding grounds with its own label construction, so it cannot be compared head-to-head with a Pakistani result. Fine-tuning a large foundation model needs more GPU memory than free tiers offer, and we found no public release of the fine-tuned locust model.

Relation to this work. It sets the reference level for the problem. We use a much smaller ConvLSTM and treat foundation-model fine-tuning only as a stretch goal.

3.2.1.5 Yusuf et al. and Barbet-Massin et al. — pseudo-absence generation [24, 22]

Summary. Yusuf et al. examine how the way pseudo-absence points are generated affects machine-learning models of locust breeding grounds in Africa. Barbet-Massin et al. compare pseudo-absence selection strategies in species distribution models and show that the preferred strategy depends on the modelling technique.

Critical analysis. Both address the presence-only problem directly, which most predictive studies treat as a preprocessing detail. The locust study covers African data and the schemes it tested; the ecological study uses static suitability models, so its lesson is methodological rather than directly transferable numbers.

Relation to this work. They motivate treating negative sampling as an experimental variable: we compare three negative-example strategies instead of choosing one, and do not assume one choice is universally best (Section 5.6.2).

3.2.2 Spatiotemporal Deep Learning for Gridded Sequences

3.2.2.1 Shi et al. — Convolutional LSTM [17]

Summary. ConvLSTM extends the LSTM [25] by replacing the matrix products in its input-to-state and state-to-state transitions with convolutions, so that hidden states are spatial maps. It was introduced for precipitation nowcasting from radar sequences, where it outperformed a fully connected LSTM and an operational optical-flow method.

Critical analysis. It captures both spatial neighbourhoods and temporal evolution, and it is small enough to train on modest hardware. It expects regular, gap-free input grids and can overfit when the number of independent sequences is small.

Relation to this work. It is the core of the locust forecaster; the regional grid is treated as an evolving image, and the small number of training sequences is a design concern (Section 5.6.4).

3.2.3 Unsupervised Anomaly Detection in Crop Monitoring

3.2.3.1 Mouret et al. — parcel-level outlier detection [4, 26]

Summary. The first study detects anomalous crop development at parcel level, without labels, on 3,361 wheat parcels (2016/2017) and 2,218 rapeseed parcels (2017/2018) in the Beauce region of France. Pixel-level indices are summarised per parcel by the median and interquartile range, assembled into a feature vector over dates, and scored by an Isolation Forest. With an outlier ratio of 10%, 94.1% of detected rapeseed parcels and 95.5% of detected wheat parcels were confirmed agronomic anomalies. A companion study reconstructs cloud-affected Sentinel-2 series with robust Gaussian mixture models before the same kind of anomaly detection.

Critical analysis. This is the best methodological precedent for our task, and its configuration (1,000 trees, subsample size 256, a 10 m inner buffer, zonal median and IQR) is simple and robust. However, its "true positives" are anomalies of any kind confirmed by experts—heterogeneity, late growth, database errors—not disease; precision rather than recall was measured; and the parcels are large French fields in one region. The reconstruction method addresses missing observations but adds modelling complexity.

Relation to this work. This is the primary basis of the crop module. We differ in using Sentinel-2 only, Pakistani smallholder fields, peer-group normalisation and a threshold calibrated on local labels, and we use simpler interpolation with a missingness indicator, keeping the reconstruction method as an upgrade if gaps prove severe.

3.2.3.2 Liu et al. — Isolation Forest [19, 27]

Summary. Isolation Forest builds an ensemble of random trees; points that are isolated after few random splits receive high anomaly scores. It needs no distance computation and has linear time complexity and low memory use.

Critical analysis. It scales well and has few parameters, but it assumes anomalies are rare and different, and irrelevant features can dilute its sensitivity. It can fail when anomalies dominate a group, as in an epidemic year.

Relation to this work. It is the primary crop model; peer-group stratification and local calibration respond to its stated weaknesses (Section 5.7.5).

3.2.3.3 Ahmad — EigenCL [28] and contrastive learning [29]

Summary. EigenCL applies unsupervised, eigenvector-guided contrastive learning, in the spirit of SimCLR [29], to Sentinel-2 red-edge (NDRE) time series from Iowa corn, and reports detecting stress up to

12 days earlier than conventional thresholding and generalising to Nebraska without retraining.

Critical analysis. It is an appealing label-free deep approach, but it is a single-author preprint validated on US corn, and we found no public code or trained weights, so it would have to be re-implemented blind and trained on a GPU.

Relation to this work. It was the core of an earlier version of the crop module and was replaced by the Isolation Forest for reproducibility (Section 5.3); it remains a candidate extension.

3.2.4 Sentinel-2 Wheat Stress and Disease Monitoring

3.2.4.1 Shi et al. and Ma et al. — supervised yellow-rust detection [13, 30]

Summary. Shi et al. use a fast Fourier convolutional network on Sentinel-2 time series to discriminate wheat yellow rust from nitrogen deficiency at sites in Shaanxi province and Shunyi (China). Ma et al. build models targeted at the jointing, heading and filling periods of wheat, using multi-temporal Sentinel-2 imagery to follow yellow rust progression through spring.

Critical analysis. Together they show that stress *type* can be separated and that stage-specific design overcomes the limits of single-date models. Both are supervised and disease-specific, need labelled ground truth collected in the field, and have no Pakistani test.

Relation to this work. They explain why our output is an alert and not a diagnosis (without comparable local labels we cannot separate stress causes) and why fields are compared only against peers of the same crop stage (Section 5.7.4).

3.2.4.2 Zhang et al. — RustMAE [31]

Summary. A masked-autoencoder spatiotemporal Transformer, pre-trained on unlabeled Sentinel-2 imagery and fine-tuned on Sentinel-2, Sentinel-1 and meteorological sequences, gives 14-day-ahead stripe rust risk predictions in three Chinese provinces, with overall accuracy 87.23%, F1 84.77% and AUC 0.924.

Critical analysis. A strong supervised early-warning result, but it relies on multi-source inputs, a large model and Chinese survey labels that are not available for Pakistan.

Relation to this work. It is an external benchmark and a possible zero-shot transfer baseline, not training data; its result does not establish Pakistani performance.

3.2.4.3 Singh et al. — yellow rust in Indian Punjab [32]

Summary. A deep artificial neural network classifies yellow rust of wheat in districts of Indian Punjab from six spectral indices (including NDVI, NDWI, MCARI, a red-edge chlorophyll index, NDMI and S2REP) computed from single cloud-free Sentinel-2 dates in early 2020.

Critical analysis. Geographically closest to Pakistani Punjab, which is encouraging, but it uses single dates, limited sites and field-labelled points, and it does not test transfer across the border.

Relation to this work. It shows that red-edge and moisture indices carry rust-related signal in the region, supporting the multiband feature set.

3.2.5 Pakistani Data Sources and Context

3.2.5.1 Ul Haq et al. — ADB ground-truthing survey [33]

Summary. A public Zenodo record of ground-truth crop-type survey data collected during the Rabi 2022–23 and Kharif 2023 seasons in Pakistan, with date, coordinates, crop code, district and crop stage. It holds roughly 137,000 survey points in total; the project's data review lists 43,892 Rabi and 92,951 Kharif records, to be confirmed on download.

Critical analysis. It is the largest open source of dated, georeferenced Pakistani crop observations, but it labels crop type and stage, not health, and covers a single season.

Relation to this work. It defines the wheat cohort (Section 5.7.2); it also means that a held-out *season* cannot be tested with this source alone.

3.2.5.2 Khan et al. — crop type in Pakistan [14, 34]

Summary. A stacked LSTM on Sentinel-2 time series identifies crop types for smallholder farms in Punjab and reports 93.77% accuracy for major crops (rice, wheat, sugarcane) even when only the first month of data is used. A public dataset from the UET Peshawar group provides ground-truth crop classes (more than ten types) with Sentinel-2 and PlanetScope imagery for a site in Khyber Pakhtunkhwa.

Critical analysis. Together they confirm that Sentinel-2 can separate crops on small Pakistani fields, but both map crop type, not health, and cover a few districts or a single site.

Relation to this work. They support the feasibility of smallholder field-level extraction and the mixed-pixel risk that the buffering rules respond to; the dataset is a supplementary wheat-location source and a test bed for the extraction code.

3.2.5.3 Rehman et al. — wheat yellow rust surveillance [3]

Summary. A four-year (2019–2022) surveillance of the Pak-Afghan bordering Hindukush region found yellow rust to be the primary wheat disease, affecting 83% of surveyed fields with 29% showing severity of 50% or more; the data description reported to us lists about 1,145 field assessments, to be confirmed on access.

Critical analysis. It is exactly the kind of local health information needed for validation, but the data are request-only, it covers a highland region, and the reported prevalence shows that the assumption "anomalies are rare" can fail in an epidemic.

Relation to this work. It is the preferred calibration source for the crop module, and its 83% figure is the reason the module compares fields with peers and does not assume a fixed contamination rate.

3.2.5.4 Waseem, Arshad and Younas et al. — drought and flood impacts [15, 16, 35]

Summary. Waseem et al. analyse drought and standardised residual yield series across Punjab; Arshad et al. relate agrometeorological droughts to winter-wheat yield-loss risk at regional scale in Pakistan using machine learning; Younas et al. use Landsat 9, land-cover and SMAP soil-moisture data to quantify the 2022 flood damage in four areas of Sindh, finding Kharif crops most affected and Rabi cultivation delayed by high post-flood soil moisture.

Critical analysis. All three work at district or regional scale with yield or land-cover data, not field-level imagery, so they cannot label individual fields; the flood study is a preprint focused on one province.

Relation to this work. They document drought and floods as major causes of abnormal wheat years and motivate treating them as confounders. The ADB wheat season (Rabi 2022–23) followed the floods, so late sowing is a realistic false-alert source and a possible label-free sanity check (Section 5.7.8).

3.2.6 Domain Shift and Super-Resolution

3.2.6.1 Nyborg et al. and Lanaras et al. — transfer and resolution [12, 36]

Summary. Nyborg et al. show that crop classifiers trained in one region perform poorly in another because of the temporal shift of crop phenology; TimeMatch estimates that shift and adapts without target labels, outperforming competing methods by 11% in average F1 across five adaptation scenarios in Europe. Lanaras et al. train a deep network that super-resolves the 20 m and 60 m Sentinel-2 bands to 10 m and generalises globally.

Critical analysis. TimeMatch addresses transfer between European regions for crop-type classification, not health, and still requires a labelled source region. A super-resolved band is an estimate, not new

measurement, and the extra step adds a dependency on a pretrained model.

Relation to this work. They support building the Pakistani cohort from Pakistani data and treating any foreign supervised model as an external baseline. Super-resolution was considered for the red-edge and SWIR bands and not adopted: for field-level medians, plain bilinear resampling to 10 m is simpler and sufficient, and the choice is documented as an assumption.

3.3 Literature Review Summary Table

Table 3.1 summarises the 25 studies and datasets reviewed above, one row per entry, in the format required for research projects.

Table 3.1: Summary of the reviewed studies

Author	Method	Results	Limitations
Klein et al. [23]; Kimathi et al. [8]	Satellite-data review; MaxEnt on 9,134 occurrence records with climate and edaphic predictors	Environmental drivers established; suitability maps for East Africa	Static suitability; no temporal dynamics or lead time
Ellenburg et al. [7]	Soil-texture and modelled soil-moisture thresholds against FAO observations, East Africa	All thresholds met matched egg-laying period 62.5% (85.2% with 3×3 buffer)	One region and upsurge; not a forecaster
Khan et al. [11]	Machine-learning classifiers on GLAD42 (FAO records + climate, 42 countries)	Swarm-prediction classifiers on a public multi-country dataset	Tabular; no spatial neighbourhoods
Yusuf et al. [10]	Environmental series and EO images; Prithvi foundation model fine-tuned	F1 = 81.53% (Africa)	African labels; large model; weights not found public
Yusuf et al. [24]; Barbet-Massin et al. [22]	Effect of pseudo-absence generation on breeding-ground and species-distribution models	Preferred scheme depends on the model technique	African data; ecology, static models
Shi et al. [17]	ConvLSTM for radar precipitation nowcasting	Outperforms FC-LSTM and an optical-flow baseline	Needs regular grids; overfits with few sequences
Mouret et al. [4, 26]	Isolation Forest on parcel zonal statistics of Sentinel-1/2, France; Gaussian-mixture gap reconstruction	At 10% outlier ratio, 95.5% (wheat) and 94.1% (rapeseed) of detections confirmed anomalous	Anomalies of any kind, not disease; precision only
Liu et al. [19]	Isolation Forest (random-split isolation)	Linear-time anomaly scores with low memory	Assumes rare anomalies
Ahmad [28]	Eigenvector-guided contrastive learning on Sentinel-2 NDRE	Stress detected up to 12 days earlier; generalises to a second US region	Preprint; US corn; no public code
Shi et al. [13]; Ma et al. [30]	Fourier CNN; stage-specific multi-temporal Sentinel-2 models	Yellow rust separated from nitrogen deficiency and tracked by growth stage	Supervised; Chinese sites; no Pakistani test
Zhang et al. [31]	MAE-pretrained spatiotemporal Transformer, Sentinel-2/1 + weather	14-day warning: OA 87.23%, F1 84.77%, AUC 0.924	Large, multi-source; Chinese labels
Singh et al. [32]	ANN on six Sentinel-2 indices, Indian Punjab	Yellow rust vs. healthy from single dates	Single dates; limited sites
Ul Haq et al. [33]	ADB ground-truth crop survey, Pakistan, Rabi 2022–23 and Kharif 2023	About 137,000 dated, georeferenced points	Crop type and stage, not health

Author	Method	Results	Limitations
Khan et al. [14, 34]	Stacked LSTM on Sentinel-2, Punjab; UET Peshawar crop dataset	93.77% accuracy for major crops from first-month data	Crop type, not health; few districts
Rehman et al. [3]	Four-year field surveillance, Hindukush region	Yellow rust in 83% of fields; 29% with severity $\geq$50%	Request-only; highland region
Waseem et al. [15]; Arshad et al. [16]; Younas et al. [35]	Drought–yield analysis, Punjab; drought yield-loss risk; 2022 flood impact, Sindh	Drought and floods linked to yield loss and delayed Rabi sowing	District or regional scale; no field labels
Nyborg et al. [12]; Lanaras et al. [36]	Temporal-shift adaptation; CNN super-resolution of Sentinel-2 bands	+11% average F1 in five scenarios; 10 m estimate of 20/60 m bands	Crop type, Europe; estimate, not measurement

3.4 Related Applications

Four existing applications are reviewed: three operated by FAO for locusts and crops, and one commercial field-monitoring platform. Table 3.2 compares them.

3.4.1 FAO eLocust3 Suite (Field Data Collection)

Summary. National survey and control officers record standardised field data on weather, ecology, locusts and control with the eLocust3 tablet or smartphone suite, which transmits them in near real time to the national locust centre [37]; the national information officer checks and imports them into RAMSES, from which they reach FAO's Desert Locust Information Service. **Critical analysis.** It is the reference standard for locust field data and the ultimate source of our labels, but it is a collection tool, not a predictor, and record coverage depends on where teams can go. **Relation to this work.** AgriWatch consumes these records; their known imperfections (survey-effort bias, duplicate or erroneous records) are the reason for the reliability filter, the three-valued labels and the spatially tolerant metrics of Section 5.6.2.

3.4.2 FAO DLIS, RAMSES/SWARMS and the Locust Hub (Analysis and Dissemination)

Summary. National information officers combine field data with satellite rainfall and vegetation imagery and historical records in RAMSES; the data then go to FAO's SWARMS global GIS for analysis, forecasting and warnings [38]. FAO issues monthly bulletins and forecasts up to six weeks ahead [6], and the Locust Hub is the central website for the latest situation [39]. **Critical analysis.** This is the authoritative operational system, but forecasting is expert-driven, and, to the extent we could find, it does not publish an open, gridded machine-learning forecast for Pakistan evaluated on the 2019–2020 events. **Relation to this work.** The Locust Risk Module is a complement, not a replacement: an open, reproducible, Pakistan-specific risk map that could help prioritise surveys.

3.4.3 FAO Agricultural Stress Index System (ASIS)

Summary. ASIS is FAO's global agricultural drought monitoring system: from ten-day METOP-AVHRR vegetation and land-surface-temperature data at 1 km it computes the Vegetation Health Index and the percentage of cropland affected by drought per administrative unit, with country-level versions including Pakistan [40, 41]. **Critical analysis.** It is open, operational and based on the same principle as our crop module (a departure from seasonal norm), but at 1 km and administrative-unit scale, aimed at drought, and unable to point at individual fields. **Relation to this work.** The crop module works at field scale with 10 m Sentinel-2 and peer comparison; ASIS is a natural context layer for the district-level confounder analysis of Section 5.7.8.

3.4.4 EOSDA Crop Monitoring (Commercial Field Monitoring)

Summary. A commercial web platform using 10 m Sentinel-2 imagery (optionally 3 m PlanetScope) that shows NDVI and other indices, ranks fields, sends notifications about index changes, supports scouting and provides weather data [42]. **Critical analysis.** It shows that field-level satellite monitoring is practical and wanted, and its field ranking and scouting are a good interface model, but it is proprietary, its algorithms and validation are not openly documented in the sources we reviewed, and it is mainly index- and threshold-driven. **Relation to this work.** AgriWatch's crop module is an open, documented alternative for Pakistani wheat: multiband, peer-normalised, with explicit evidence tiers and honest limits; the dashboard borrows the ranked-list idea.

Table 3.2: Summary of related applications

Application	Features	Relevance	Limitations
eLocust3 suite [37]	Standardised field data on locusts, ecology and control; satellite and smartphone transmission to national centres	Source of the observation records used as labels	Collection tool only; coverage and quality depend on survey effort; erroneous records occur [43]
FAO DLIS, RAMSES/SWARMS, Locust Hub [38, 6, 39]	National and global GIS analysis; monthly bulletins; forecasts up to six weeks; central data site	Authoritative operational reference for the locust situation	Expert-driven; no open gridded learned forecast for Pakistan found
FAO ASIS [40, 41]	10-day AVHRR vegetation health index; cropland share affected by drought per administrative unit; country-level versions incl. Pakistan	Same principle (departure from seasonal norm) for crops	1 km, administrative scale; drought only; no field-level output
EOSDA Crop Monitoring [42]	Sentinel-2 field NDVI and other indices; field ranking; notifications; scouting; weather	Shows demand for field-level monitoring; interface ideas	Proprietary; methods and validation not openly documented; index/threshold driven

3.5 Research Gap

Three gaps follow from the review. (i) No open Pakistan-specific locust dataset and learned forecaster validated on a Pakistani invasion was found, and the presence-only label problem is rarely treated as an experimental variable [24]. (ii) Label-free parcel anomaly detection has strong precedent in France [4] but has not been applied to Pakistani smallholder wheat, where crop location is known but health labels are not. (iii) Evidence tiers are seldom made explicit: supervised accuracy and anomaly ranking are often compared as though equivalent.

3.6 Conclusion

The literature shows that soil moisture, vegetation and climate carry breeding signal [7, 8, 23]; that ConvLSTM is a suitable and compact model for gridded sequences [17]; that Isolation Forest on parcel-level Sentinel features can rank anomalous crop development without labels [4]; and that stress *type* and calibrated accuracy need local labels that the open Pakistani data do not provide [13, 33, 3]. These findings fix the design: a forecaster trained on a Pakistani locust dataset, a label-free crop scorer calibrated with local evidence, and an evaluation that separates ranking from classification evidence.

Chapter 4 Software Requirement Specifications

This chapter specifies what AgriWatch must do and under what conditions, for *both* of its modules: the Locust Risk Module and the Crop Abnormality Screening Module, together with the shared evaluation harness and dashboard. AgriWatch is an R&D prototype, so the "users" are the research team, domain reviewers and evaluators, and the requirements concentrate on data handling, modelling, evaluation and presentation of results, not on accounts or transactions. Requirements are numbered so that Chapter 5 (methodology) and Chapter 6 (design) can trace to them.

4.1 List of Features

Table 4.1 lists the features of the system and the module each belongs to.

Table 4.1: Features of AgriWatch

ID	Feature	Module
F1	Registration of open data sources with a provenance manifest (source, licence, access date, coverage, checksum)	Shared
F2	Selection and cleaning of FAO locust observation records for the study region	Locust
F3	Gridded anomaly cube of five input channels (plus an observed-mask channel) at 0.25° and 16-day resolution	Locust
F4	Masked three-valued locust labels at three horizons, with three negative-example strategies	Locust
F5	Baseline ladder (B0–B3, plus a recent-sightings baseline) and ConvLSTM forecaster, evaluated by rolling-origin forecasts over a real invasion	Locust
F6	Locust risk maps per forecast origin and horizon	Locust
F7	Wheat cohort construction from public crop-ground-truth data with logged exclusions	Crop
F8	Sentinel-2 cloud masking, 10-day composites, five indices and temporal features per field–season	Crop
F9	Peer-group normalisation and Isolation Forest anomaly scoring, including early-season (as-of-date) scoring	Crop
F10	Threshold calibration against verified Pakistani observations and evidence-tier labelling	Crop
F11	Leakage-controlled evaluation harness (splits, embargo, metrics, bootstrap, seeds)	Shared
F12	Dashboard showing locust risk maps and ranked wheat-field alerts with the evidence level and limitations	Shared
F13	Export of tables and figures together with configuration and manifest	Shared

4.2 Functional Requirements

The *actors* are: the **Analyst** (a team member who runs pipelines and experiments), the **Domain reviewer** (a locust officer, agronomist or plant pathologist who inspects outputs and may supply verified field observations) and the **Evaluator** (supervisor or panel who inspects results and provenance).

Table 4.2: Functional requirements

ID	Requirement	Module	Actor
FR-01	The system shall record every data source in a manifest with source identifier, licence, access date, coverage, coordinate system and checksum, and shall store the acknowledgement text required by the licence.	Shared	Analyst
FR-02	The system shall import FAO observation records and select them by coordinates inside the configured study box; the country-code and country-name columns shall never be used for selection (country name is used only to report counts).	Locust	Analyst
FR-03	The system shall retain only records with valid coordinates and observation dates, shall treat records of category NO LOCUST only as survey-confirmed absences, and shall keep them out of the positive set.	Locust	Analyst
FR-04	The system shall build an anomaly cube of shape (time, channel, rows, columns) for the five anomaly channels, with climatology fitted on 2001–2019 (training years only), and a mask channel for missing values.	Locust	Analyst
FR-05	The system shall generate label tensors Y_h and masks M_h for horizons 28, 42 and 56 days using target windows defined by calendar date, such that no input overlaps a target window.	Locust	Analyst
FR-06	The system shall support three negative-example strategies (survey-confirmed absences, stratified pseudo-absences, random pseudo-absences), selectable by configuration.	Locust	Analyst
FR-07	The system shall train and evaluate baselines B0–B3, a recent-sightings baseline and the ConvLSTM on frozen time-blocked splits with a per-horizon embargo and by rolling-origin forecasts, using several random seeds.	Locust	Analyst
FR-08	The system shall output a risk map (probability per cell) for every forecast origin and horizon.	Locust	Analyst, Domain reviewer
FR-09	The system shall build the wheat cohort from the ADB survey applying the inclusion and exclusion rules, and shall store the reason for every exclusion.	Crop	Analyst
FR-10	The system shall extract cloud-masked 10-day composites, five indices and temporal features per field–season, using an inward-buffered polygon where available and a 3×3 window (flagged point-derived) otherwise.	Crop	Analyst
FR-11	The system shall form peer groups by district, season and crop stage and shall peer-normalise features using training statistics only.	Crop	Analyst
FR-12	The system shall fit an Isolation Forest and output a continuous anomaly score per field.	Crop	Analyst
FR-13	The system shall select a binary threshold only when independently verified Normal/Abnormal observations exist, and otherwise shall output the ranking only.	Crop	Analyst, Domain reviewer
FR-14	The system shall support as-of-date scoring that uses only composites up to a given date.	Crop	Analyst
FR-15	The evaluation harness shall be the only component that creates splits (locust module: by time; crop module: by district); it shall verify the absence of leakage and abort a run if any check fails.	Shared	Analyst
FR-16	The system shall run the planned experiments and store each result together with the configuration, seeds and manifest identifiers.	Shared	Analyst

ID	Requirement	Module	Actor
FR-17	The dashboard shall display locust risk maps by horizon and forecast origin, with an overlay of observed records, and ranked wheat-field alerts by district and stage.	Shared	Domain reviewer, Evaluator
FR-18	The dashboard shall state, for every displayed layer, its evidence tier and its limitations.	Shared	All
FR-19	In public mode the dashboard shall show crop results only in aggregated or anonymised form.	Crop	All
FR-20	The system shall let a Domain reviewer record the outcome of a field inspection as a candidate verified label with provenance.	Crop	Domain reviewer
FR-21	The system shall export tables and figures with the configuration and manifest that produced them.	Shared	Analyst

4.3 Quality Attributes

- **Reproducibility.** Any reported number or figure can be regenerated from versioned code, a configuration file, the manifest and fixed seeds.
- **Correctness (leak-free evaluation).** Splits, climatology, peer statistics and thresholds are fitted without test data; automated checks enforce this.
- **Interpretability.** Features are named indices and temporal summaries, so an alert can be traced to the variables that caused it.
- **Traceability.** Every output is linked to its sources, licences and the requirement it fulfils.
- **Modularity and maintainability.** The locust and crop pipelines share only the index library and the evaluation harness and can be changed independently.
- **Usability.** The dashboard states what each layer means and how much to trust it, in plain language, for users who are not modellers.
- **Reliability.** Long exports are restartable and cached; a failed export does not corrupt earlier results.

4.4 Non-Functional Requirements

Table 4.3: Non-functional requirements

ID	Category	Requirement
NFR-01	Performance (feasibility)	The data audit shall run on a laptop in minutes; the ConvLSTM ($\approx 2.7 \times 10^5$ parameters) shall be trainable on a free-tier GPU; 20-year exports shall use server-side reduction so that only small arrays are downloaded.
NFR-02	Scalability	The study box, cell size, horizons, season and crop code shall be configuration values, so that another region or crop needs no code change.

ID	Category	Requirement
NFR-03	Reproducibility	Five random seeds (42, 52, 62, 72, 82) shall be used for stochastic models; software versions and scene identifiers shall be logged.
NFR-04	Security and privacy	Survey points identify private farms: public outputs shall be aggregated or anonymised; credentials and tokens shall never be stored in the repository.
NFR-05	Legal and compliance	Each data source's licence shall be honoured; FAO shall be acknowledged and the project kept non-commercial; the manifest shall hold the licence text.
NFR-06	Portability	The pipelines shall run on common operating systems with open-source Python tools; the report shall compile on Overleaf.
NFR-07	Availability	The system is batch-oriented; no continuous (24/7) service or real-time operation is required.
NFR-08	Reusability	The constructed locust dataset and the wheat cohort manifest shall be documented well enough for release and reuse.
NFR-09	Extensibility	New baselines or models shall be addable through a common interface (fit, predict, save) without changing the evaluation harness.

4.5 Assumptions

- The FAO observation data remain available under the stated licence through the open catalogue, which supplied the multi-year records [44, 39].
- Earth Engine access for research stays free, and Sentinel-2 Level-2A covers the Rabi 2022–23 season [45, 9].
- The ADB survey stays downloadable [33]; verified wheat-health observations can be obtained by request [3]—if not, the crop module's claims are limited to anomaly ranking.
- The team has no field access: no new field, drone or hyperspectral data are collected.
- Outputs are read by people with domain knowledge and are treated as risk or inspection priorities, not diagnoses.

4.6 Use Cases

Seven use cases cover the system's central functions across both modules. The tables follow the standard format (name, actors, summary, conditions, special requirements, basic and alternative flows); UC2–UC4 concern the Locust Risk Module, UC5–UC7 the Crop Abnormality Screening Module, and UC1 is shared.

Table 4.4: Use case UC1: register and verify a data source

Name	UC1 — Register and verify a data source		
Actors	Analyst		
Summary	The analyst records a source with its licence, coverage and checksum before any pipeline may use it.		
Pre-Conditions	The source is reachable (or its access request has been submitted); the analyst can read its licence.		
Post-Conditions	A manifest entry exists with licence text and checksum, and the source is marked verified, pending or blocked.		
Special Requirements	The licence and acknowledgement text must be stored; non-commercial or request-only conditions must be flagged (FR-01, NFR-05).		
Basic Flow			
Step	**Actor Action**	**Step**	**System Response**
1	Provides the source identifier, location and access date.	2	Reads the metadata, computes a checksum and stores a manifest entry.
3	Reviews the licence, coverage and coordinate system.	4	Marks the source *verified* and makes it available to the pipelines.
Alternative Flow			
3-A	Access requires approval or the file cannot be read.	4-A	Marks the source *pending* or *blocked* and prevents dependent steps from running.

Table 4.5: Use case UC2: build the locust dataset

Name	UC2 — Build the locust dataset		
Actors	Analyst		
Summary	The analyst builds the anomaly cube and the masked labels for the study region from FAO records and the five channels.		
Pre-Conditions	FAO records and channel sources are verified (UC1); the configuration (box, grid, horizons, split years) is set.		
Post-Conditions	The cube, mask cube, label tensors, masks, splits and per-year counts of positives are stored with their configuration.		
Special Requirements	Climatology uses training years only (2001–2019); records are selected by coordinates (FR-02 to FR-06).		
Basic Flow			
Step	**Actor Action**	**Step**	**System Response**
1	Starts the locust build with a configuration.	2	Selects the FAO records inside the study box by coordinates and drops invalid ones.
		3	Builds the five channels per cell and 16-day step and fits climatology on training years.
4	Chooses a negative-example strategy.	5	Builds Y_h and M_h and reports positives per year and horizon.
6	Reviews the counts.	7	Freezes the splits and stores all artefacts.
Alternative Flow			
6-A	The training window holds too few positives.	7-A	Shows a data-sufficiency warning and offers the wider-region or rolling-origin option.

4.7 Hardware and Software Requirements

4.7.1 Hardware Requirements

A laptop-class computer is sufficient for data handling, the audit, the classical baselines and the Isolation Forest. The ConvLSTM ($\approx 2.7 \times 10^5$ parameters) is trained on a free-tier cloud GPU; no cluster is needed. Storage must hold the exported arrays (small, because reduction is done server-side) and the manifest.

Table 4.6: Use case UC3: train and evaluate the locust models

Name	UC3 — Train and evaluate the locust models		
Actors	Analyst		
Summary	The analyst trains baselines B0–B3, the recent-sightings baseline and the ConvLSTM and evaluates them by rolling-origin forecasts over the 2019–2020 invasion.		
Pre-Conditions	UC2 is complete and the splits are frozen.		
Post-Conditions	Metrics with intervals, thresholds, seeds and configuration are stored for every model and horizon.		
Special Requirements	Each forecast origin is scored once, by a model fitted only on labels known at that origin; all leakage checks must pass first (FR-07, FR-15, FR-16).		
Basic Flow			
Step	**Actor Action**	**Step**	**System Response**
1	Selects models and seeds.	2	Runs the leakage checks on the splits.
		3	Trains each model with early stopping on validation PR-AUC.
4	Confirms the frozen thresholds.	5	Scores every rolling-origin forecast once and stores the metrics.
6	Compares the ConvLSTM with the baselines against the success criterion.	7	Records whether an advantage was demonstrated.
Alternative Flow			
2-A	A leakage check fails.	3-A	Aborts the run and reports the violation.

Table 4.7: Use case UC4: view a locust risk map

Name	UC4 — View a locust risk map		
Actors	Domain reviewer, Analyst, Evaluator		
Summary	The user views the predicted risk for a horizon and forecast origin, with observed records and the evidence label.		
Pre-Conditions	At least one model run exists in the artefact store.		
Post-Conditions	None (read-only).		
Special Requirements	The evidence tier and limitations must be shown (FR-08, FR-17, FR-18).		
Basic Flow			
Step	**Actor Action**	**Step**	**System Response**
1	Selects a horizon (28, 42 or 56 days) and a forecast origin.	2	Renders the risk map for the study region.
3	Switches on the overlay of observed records.	4	Draws the records and shows the evidence tier and limitations.
Alternative Flow			
1-A	No run exists for the chosen origin.	2-A	Shows a message and lists the available origins.

4.7.2 Software Requirements

4.8 Graphical User Interface

The dashboard is a read-only interface over the artefact store. Three screens cover the use cases; their navigation is in Figure 4.1, and Table 4.12 states which use cases and requirements each screen covers.

4.8.1 Screen S1: Overview Map

Screen S1 (Figure 4.2) is the landing screen. It shows the study-region map with a selectable layer (locust risk at 28, 42 or 56 days, or wheat-field alerts), a time slider for the forecast origin, a ranked field table, and a panel that states the evidence tier and limitations of the displayed layer. It covers UC4.

Table 4.8: Use case UC5: build the wheat cohort and features

Name	UC5 — Build the wheat cohort and features
Actors	Analyst
Summary	The analyst selects wheat observations, extracts Sentinel-2 series and derives features and peer groups.
Pre-Conditions	The ADB survey and Sentinel-2 are verified (UC1); the configuration is set.
Post-Conditions	A cohort manifest with exclusion reasons and a feature table with peer groups are stored.
Special Requirements	Partitions are made by district (all fields of a district stay together); point-derived records are flagged (FR-09 to FR-11).

Basic Flow			
Step	**Actor Action**	**Step**	**System Response**
1	Starts the cohort build.	2	Selects wheat records, applies the inclusion rules and logs every exclusion.
		3	Masks clouds, builds 10-day composites, computes indices and temporal features.
		4	Forms peer groups and reports their sizes.
5	Reviews the exclusion log and group sizes.	6	Freezes the partitions.
Alternative Flow			
3-A	A field has less than 60% valid temporal coverage.	4-A	Excludes the field and logs the reason.

Table 4.9: Use case UC6: score fields and calibrate the threshold

Name	UC6 — Score fields and calibrate the threshold
Actors	Analyst, Domain reviewer (supplies labels)
Summary	The analyst fits the Isolation Forest, scores all fields and, if verified labels exist, calibrates a Normal/Abnormal threshold.
Pre-Conditions	UC5 is complete.
Post-Conditions	Anomaly scores are stored; a frozen threshold exists only if labelled validation fields were available; the evidence tier is recorded.
Special Requirements	A fixed contamination rate is not used; the threshold is chosen on validation data only (FR-12, FR-13).

Basic Flow			
Step	**Actor Action**	**Step**	**System Response**
1	Starts scoring.	2	Fits peer statistics and the forest on the training partition.
		3	Scores all fields.
4	Supplies verified Normal/Abnormal observations.	5	Chooses the threshold by maximum F1 on validation and freezes it.
		6	Evaluates once on the test partition and records the evidence tier.
Alternative Flow			
4-A	No verified labels are available.	5-A	Outputs the ranking only (and Precision@k on any inspected fields); evidence tier 1.

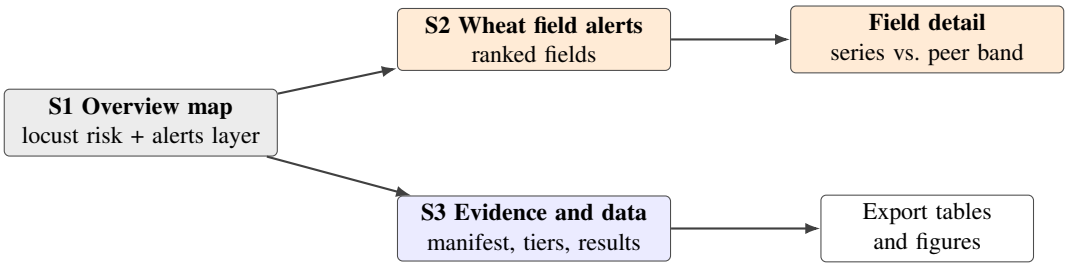


Figure 4.1: Navigation flow of the dashboard (every screen also links back to S1)

Table 4.10: Use case UC7: review ranked field alerts

Name	UC7 — Review ranked field alerts
Actors	Domain reviewer (agronomist or plant pathologist)
Summary	The reviewer examines the ranked fields of a district and stage to decide which to inspect first, and may record an inspection result.
Pre-Conditions	Scores exist (UC6).
Post-Conditions	Optionally, a candidate verified label with provenance is stored for later calibration.
Special Requirements	Public mode shows aggregated or anonymised results only; an alert is not a diagnosis (FR-17 to FR-20).

Basic Flow			
Step	**Actor Action**	**Step**	**System Response**
1	Selects a district, season and stage.	2	Lists fields by score with the features that contribute most.
3	Opens a field.	4	Plots its time series against the peer median band.
5	Records the inspection outcome (optional).	6	Stores it as a candidate label with provenance.
Alternative Flow			
1-A	The system is in public mode.	2-A	Shows district-level summaries without field identifiers.

Table 4.11: Software and service requirements

Item	**Requirement**
Cloud services	Google Earth Engine account for server-side reductions [45]; Copernicus and NASA Earthdata access for data products
Programming	Python 3 with pandas, NumPy, matplotlib, scikit-learn [46], LightGBM (gradient-boosted trees), a deep-learning framework (ConvLSTM) and geospatial libraries for raster and vector data
Data store	Files for arrays and tables, plus a small relational manifest (for example SQLite)
Dashboard	A lightweight web or notebook interface that reads the artefact store
Documentation	LaTeX (Overleaf-compatible template) and version control for code and configuration

Table 4.12: Screens, use cases and requirements

Screen	**Use cases**	**Requirements**
S1 Overview map	UC4	FR-08, FR-17, FR-18
S2 Wheat field alerts	UC7 (UC6 results)	FR-12, FR-13, FR-17 to FR-20
S3 Evidence and data	UC1 (view), UC3, UC6	FR-01, FR-16, FR-18, FR-21

4.8.2 Screen S2: Wheat Field Alerts

Screen S2 (Figure 4.3) lists the fields of a chosen district, season and stage, ranked by anomaly score, with the features contributing most; opening a field plots its index series against the peer median band. In public mode only district summaries appear. It covers UC7 and shows the results of UC6.

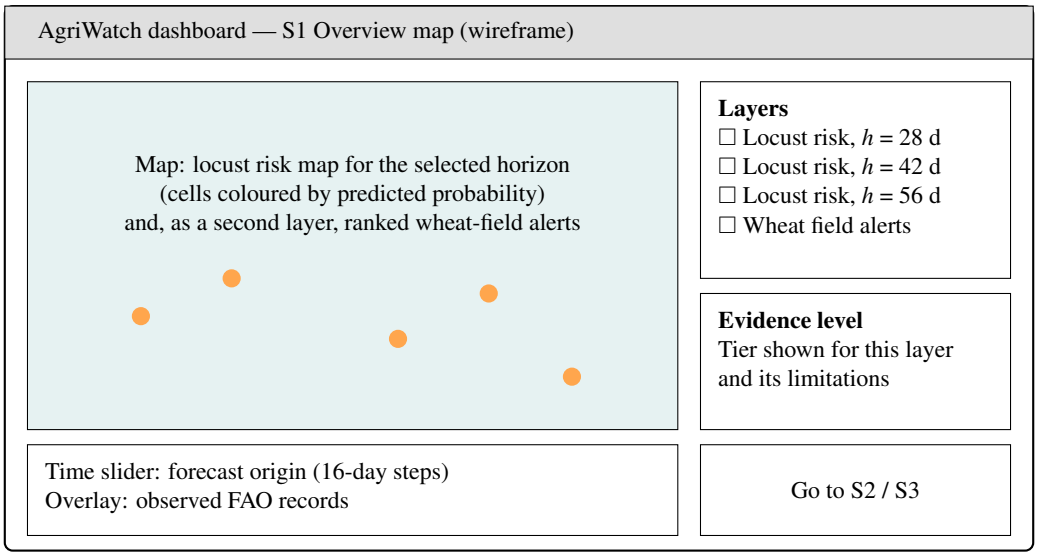


Figure 4.2: Screen S1: overview map (wireframe, not yet implemented)

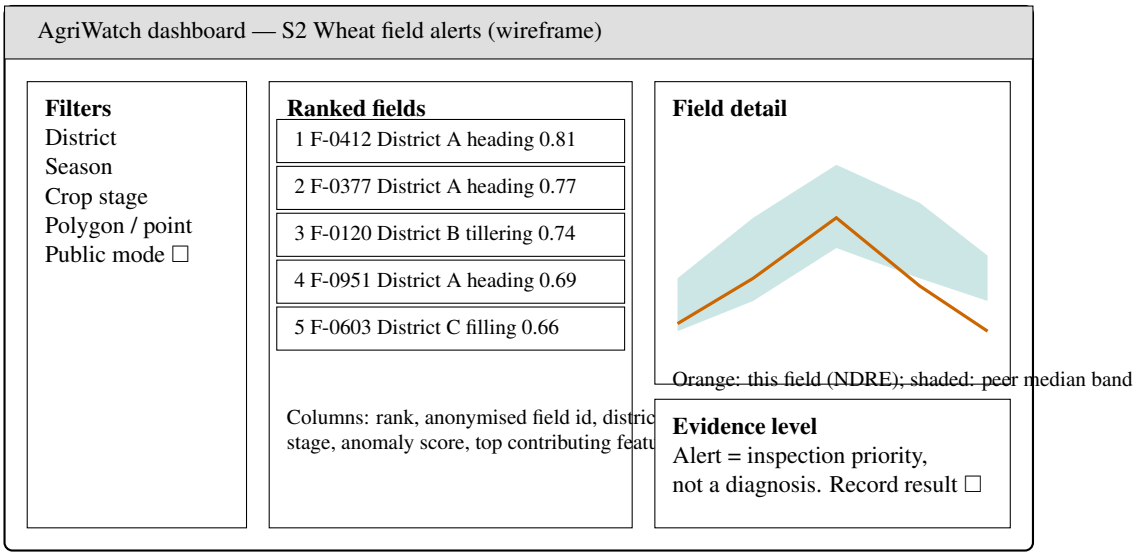


Figure 4.3: Screen S2: wheat field alerts (wireframe, not yet implemented)

4.8.3 Screen S3: Evidence and Data

Screen S3 (Figure 4.4) shows provenance and results: the source manifest with licence and status, the evidence tier reached by each module, and the experiment results with their configuration, with export of tables and figures. It covers UC1 (viewing), UC3 and UC6 results, and FR-21.

4.9 Database Design

AgriWatch does not need a transactional database. Large arrays (the anomaly cube, label tensors, composites) are stored as files, and a small relational *manifest* (for example SQLite) records provenance, cohort membership, features, runs and metrics. The conceptual entity–relationship model in Figure 4.5

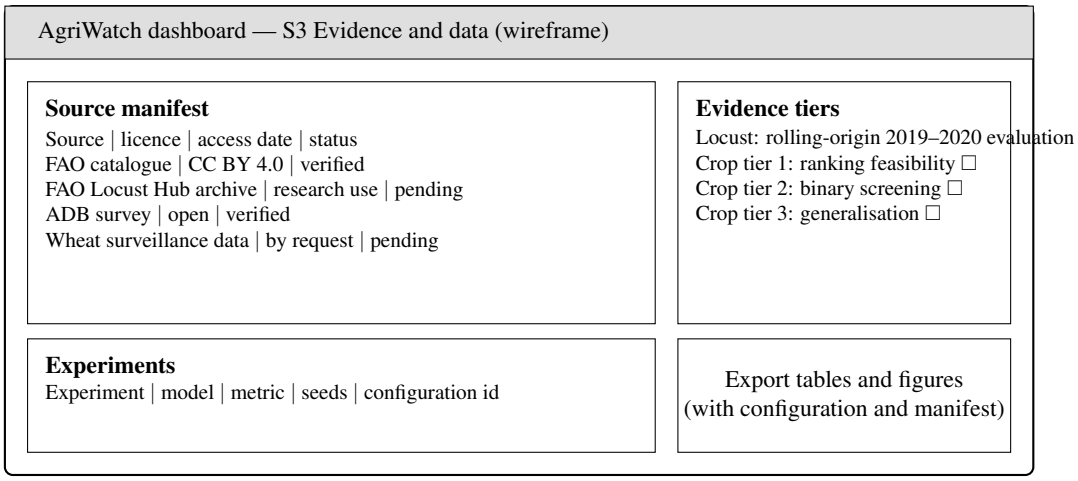


Figure 4.4: Screen S3: evidence and data (wireframe, not yet implemented)

covers both modules; the locust-specific entities are *FAO_RECORD*, *GRID_CELL* and *LABEL_CELL*, the crop-specific entities are *FIELD_SEASON* and *FEATURE_ROW*, and the rest are shared.

4.9.1 ER Diagram

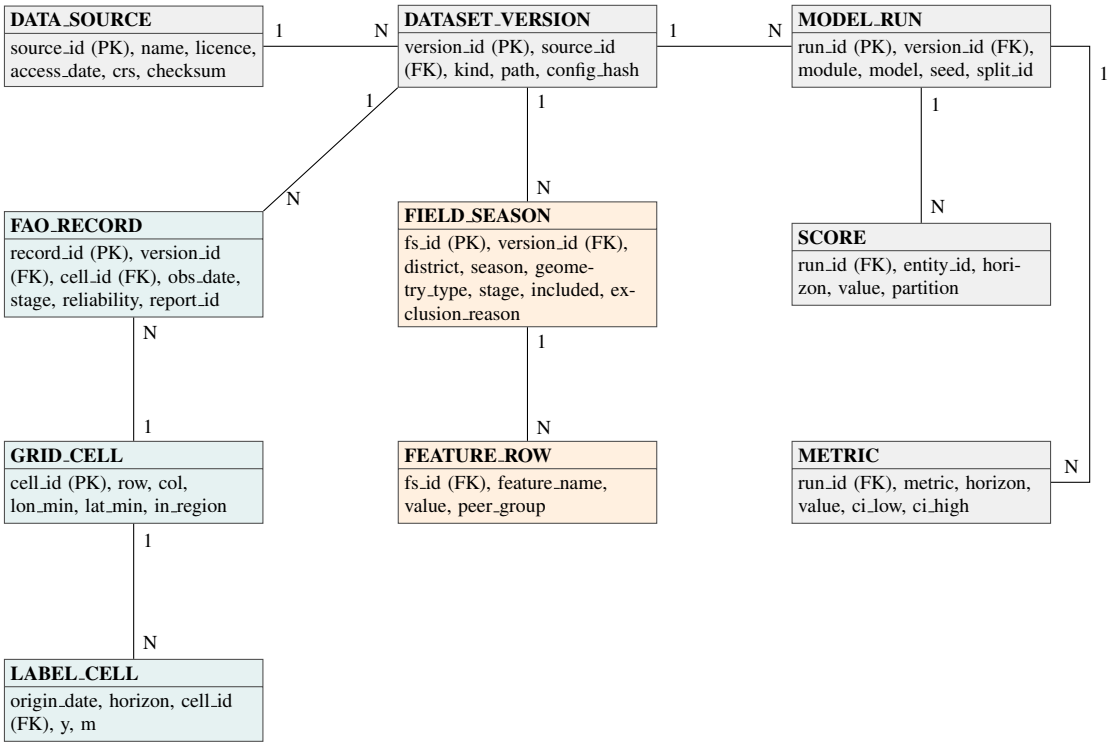


Figure 4.5: Entity–relationship model of the manifest (1 = one, N = many; teal: locust, orange: crop, grey: shared)

4.9.2 Data Dictionary

Table 4.13: Data dictionary of the manifest

Entity	Attribute	Type	Description
DATA_SOURCE	source_id	text (PK)	Unique identifier of the source
	licence	text	Licence name and acknowledgement text
	access_date	date	Date the source was accessed
	crs, checksum	text	Coordinate system and file checksum
DATASET_VERSION	version_id	text (PK)	Version of a derived or raw dataset
	kind	text	raw, cube, labels, cohort, features
	config_hash	text	Hash of the configuration that produced it
FAO_RECORD	record_id	integer (PK)	Identifier of the cleaned observation
	obs_date	date	Observation date
	stage	text	adult, hopper, band or swarm
	reliability, report_id	text	Location reliability; survey-report identifier
GRID_CELL	cell_id	integer (PK)	Cell of the 28×24 grid
	row, col	integer	Position in the grid
	in_region	boolean	Inside the region mask
LABEL_CELL	origin_date, horizon	date, integer	Forecast origin and horizon in days
	y, m	integer	Label (0/1) and mask (known = 1)
FIELD_SEASON	fs_id	text (PK)	Field–season unit (assigned to a partition through its district)
	district, season, stage	text	Peer-group keys
	geometry_type	text	polygon or point-derived
	included, exclusion_reason	boolean, text	Inclusion decision and reason
FEATURE_ROW	feature_name, value	text, real	One temporal feature of a field–season
	peer_group	text	Peer group used for normalisation
MODEL_RUN	run_id	text (PK)	One training or scoring run
	module, model	text	locust/crop; B0–B3, ConvLSTM, IF
	seed, split_id	integer, text	Random seed and split used
SCORE	entity_id	text	cell_id or fs_id
	horizon, value, partition	integer, real, text	Horizon (locust), score or probability, partition
METRIC	metric, horizon	text, integer	PR-AUC, F1, Precision@k, etc.
	value, ci_low, ci_high	real	Estimate and block-bootstrap interval

Array data (the anomaly cube of shape time $\times$ channel $\times$ 28 $\times$ 24, its mask, and the label tensors) are stored as files with a sidecar description, and are referenced from *DATASET_VERSION*.

4.10 Risk Analysis

Risks fall into four groups: data risks (R1, R2, R3, R5, R7, R8), modelling risks (R4, R9, R10, R13), project risks (R6, R11) and compliance risks (R12). Table 4.14 gives the consequence and mitigation of each; it applies to both modules.

Table 4.14: Risk register

ID	Risk	Impact	Mitigation / contingency
R1	Multi-year FAO archive cannot be retrieved through the open catalogue	High (resolved)	The catalogue's query endpoint returned the whole master view (2004–2026), so the Locust Hub request is no longer needed (Section 5.5.4); the fallbacks (coarser GLAD42-based reproduction, digitised situation maps) were not required

ID	Risk	Impact	Mitigation / contingency
R2	Few breeding-stage positives in 2008–2018 inside the study box	High (materialised)	Counted after retrieval: only 63 hopper or band cell-steps. The primary target was widened to presence at any life stage, breeding stage is reported as a secondary target, and a rolling-origin evaluation is used; the wider Southwest Asian training area remains a fallback
R3	Survey-effort bias: records reflect where surveyors went	Medium	Use "NO LOCUST" absences where available, mask unknowns, report spatially tolerant metrics, discuss as a limitation
R4	Too few independent training sequences for a deep model	Medium	Small network, early stopping, spatial crops, strong baselines; report negative results
R5	Data-product coverage gaps over 2001–2020 (for example reanalysis via Earth Engine)	Medium (materialised)	ESA CCI soil moisture is not in the Earth Engine catalogue and the ERA5 daily collection there ends on 9 July 2020; ERA5-Land (daily, complete to the end of 2020) now supplies soil moisture, temperature and rainfall
R6	Earth Engine quotas and time for 20-year exports	Medium	Server-side reductions, batch exports, caching of intermediate arrays
R7	No locally verified wheat-health labels	High	Request the surveillance data now; otherwise report ranking and Precision@k on inspected fields only; make no accuracy claims
R8	Surveillance region and years differ from the ADB season and districts	Medium	Build peer groups from observations in the same district and season (an independent winter-cereal mask can supply peers where the survey has none); decide in FYP-II
R9	Mixed pixels on smallholder fields	Medium	Inward buffering, 3×3 medians for points, minimum valid-pixel rules; report point and polygon results separately
R10	Late sowing, flood effects, senescence and epidemic prevalence confound the score	Medium	Stage-matched peers, explicit confounder analysis, no fixed contamination rate, local calibration
R11	Time and team capacity (three members, one semester of work remaining)	Medium	Baselines first; fixed scope fence (no foundation-model fine-tuning or contrastive model unless time remains)
R12	Licence and acknowledgement obligations for FAO data	Low	Cite and acknowledge FAO as required; keep the project non-commercial; share a copy of outputs with FAO as agreed
R13	Non-stationarity of locust records: models using environment alone do not beat chance in 2019–2020 (quiet years differ from the upsurge; control operations and survey effort change what is recorded)	High	Rolling-origin evaluation, a recent-sightings baseline, per-period reporting, and judging the ConvLSTM against the strongest baseline; report negative results as they stand

4.11 Conclusion

This chapter specified AgriWatch as a single system with two modules: 21 functional and 9 non-functional requirements, seven use cases (two modules plus shared provenance), three dashboard screens, a manifest data model and a risk register. The next chapter describes the methodology that fulfils these requirements.

Chapter 5 Proposed Approach and Methodology

This chapter specifies how AgriWatch will be built and evaluated. It first presents the common framework and the design decisions that shaped it, then describes the two modules in turn—data, labels, models, splits and metrics—and closes with a risk analysis. The approach is an *algorithmic and procedural* design (a framework combined with two models), not a software specification; software structure is given in Chapter 6.

5.1 Overall Framework

Both modules rest on one principle: *compare a place or field with what is expected for its season and flag the departure*. They apply it with different tools because the two problems have different shapes. Locust risk is a *forecasting* question with sparse positive labels, which a supervised spatiotemporal model suits. Wheat abnormality is a *similarity* question with no health labels, which an unsupervised anomaly scorer suits. Forcing one architecture on both would solve neither problem well. Table 5.1 contrasts the two; Figure 5.1 shows the framework.

Table 5.1: The two modules at a glance

	Locust Risk Module	**Crop Abnormality Screening**
Question	Will locusts (any life stage; breeding stage as a secondary target) be recorded in this cell 28, 42 or 56 days from now?	Is this wheat field developing abnormally compared with comparable fields?
Scale	Regional grid, 0.25° cells, 16-day steps	Field, 10-day composites over one season
Labels	FAO presence records and survey-confirmed absences (self-built dataset)	None for training; local labels only for calibration
Model	ConvLSTM with three horizon heads (plus baselines)	Isolation Forest on robust temporal features (plus baselines)
Validation	Rolling-origin forecasts over the 2019–2020 invasion	Held-out districts (never held-out seasons); verified Pakistani observations
Output	Risk maps (probability per cell)	Anomaly score; Normal/Abnormal only after calibration

5.2 Conventions Used in this Report

Four things recur throughout this report and differ between the two modules, which are separate studies: how the study region is selected, which years are used, how the data are split, and how the features are counted. Each is fixed once here *for each module*, and every later mention follows Table 5.2.

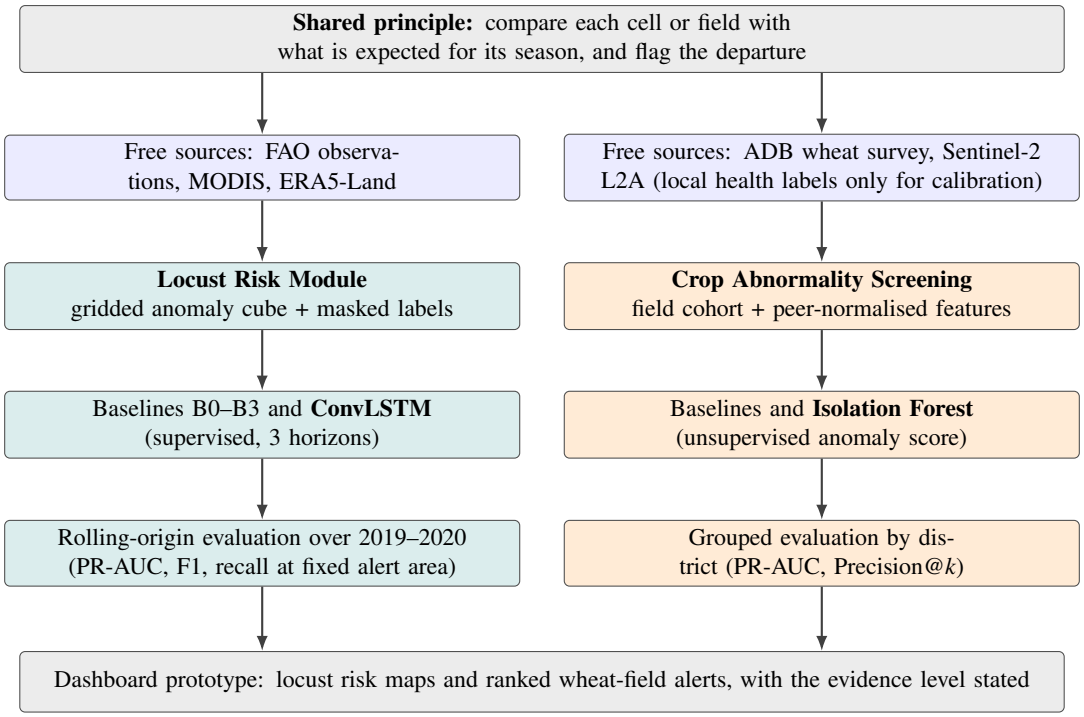


Figure 5.1: AgriWatch framework: one principle, two pipelines, one dashboard

Where the report says "training years" for the locust module it means 2008–2019. The range 2001–2019 is only the reference period of the climatology, and it starts in 2001 because MOD13Q1 begins in February 2000, so 2001 is the first complete year.

5.3 Evolution of the Design

The design in this report differs from the version first proposed. The changes were deliberate, each answers a concrete obstacle, and they are recorded here so the reader can see why the current design is what it is (Table 5.3).

Table 5.3: Design decisions that changed during FYP-I

Aspect	Change	Reason
Crop sensing	Leaf-level hyperspectral imaging → Sentinel-2 satellite imagery	The team has no hyperspectral camera or field access; existing hyperspectral datasets were small, single-timepoint and not Pakistani. Satellite data are free, repeated and cover Pakistan, and also put both modules on the same kind of data.
Crop model	Deep contrastive model (EigenCL-style) → Isolation Forest	EigenCL has no public code to check an implementation against, needs GPU tuning, and was validated on US corn [28]. Isolation Forest has a parcel-level wheat precedent [4] and runs on a laptop.

Aspect	Change	Reason
Band alignment	Pretrained super-resolution of 20 m bands → bilinear resampling	Field-level medians do not need sharper pixels, and super-resolution adds a pretrained dependency [36]. The limitation (no new spatial information) is documented.
Crop validation	Documented drought years as a label-free check → locally verified Normal/Abnormal observations	Drought records are district-scale and the public wheat survey covers a single season; binary claims require field-level evidence [15, 3].
Locust anomaly baseline	20-year climatology (2001–2020) → climatology from training years only (2001–2019)	A baseline that includes 2020 would leak evaluation-period information into the inputs.
Locust input geometry	Per-cell 32×32 patches → whole region as one 28×24 grid	At 0.25° per cell a 32×32 window is larger than the whole study region; treating the region as an evolving image is the standard ConvLSTM use [17].
Locust labels	"Presence" and random negatives → breeding-stage targets with masked unknowns and three negative strategies → presence at any life stage as the primary target (breeding stage secondary), survey-confirmed absences as the primary negatives, and two pseudo-absence strategies kept as ablations	FAO records are presence-only; unknown cells must not be treated as negatives [24]. Only 63 hopper or band cell-steps exist in 2008–2018 (Section 5.5.4), too few to train or evaluate on.
Locust data access	Locust Hub request form → the catalogue's SQL query endpoint	The endpoint returned the whole master view (2004–2026) without waiting for the request, which is still pending.
Locust weather and soil moisture	ESA CCI soil moisture and ERA5 → ERA5-Land for soil moisture, temperature and rainfall	ESA CCI is not in the Earth Engine catalogue and the ERA5 daily collection there ends on 9 July 2020; ERA5-Land is complete to the end of 2020 [18].
Locust splits	Three multi-year blocks with the 2019–2020 upsurge as the only test → train 2008–2019, validation Jan–Jun 2020, test Jul–Dec 2020, plus rolling-origin evaluation over 2019–2020	FAO records in the box begin in 2007, the upsurge starts in 2019, and 10–11 origins per horizon make a fixed validation or test block too thin; rolling-origin forecasts give 45 scored origins per horizon.

5.4 Study Region and Spatial Reference

Locust study region. The Cholistan (Bahawalpur, Bahawalnagar, Rahim Yar Khan) and Tharparkar desert-margin zones, represented by the bounding box 24–31°N, 68–74°E, divided into square cells of 0.25° (28 rows × 24 columns = 672 cells; each cell is roughly 28 km × 25 km, about 690 km^2)

Table 5.2: Region selection, years, splits and feature counts, by module

	Locust Risk Module	**Crop Abnormality Screening Module**
Region selection	By *coordinates only*: FAO records inside the box 24–31°N, 68–74°E are kept, whatever their country. The country-code and country-name columns are never used to select (the column labelled "ISO2" holds FAO legacy codes: Pakistan is `PA`, not `PK`). The box also covers part of Indian Rajasthan.	By *crop code*: wheat records of the Pakistan-wide ADB survey. There is no coordinate box; a field's coordinates only place it in a district and a Sentinel-2 tile.
Years	Satellite and climate inputs 2001–2020 (2001 is the first full MODIS year). Climatology 2001–2019. FAO records in the box begin in August 2007, so forecast targets are used from 2008. *Training targets 2008–2019*; evaluation 2019–2020.	One season only: Rabi 2022–23 (sowing October–November 2022, harvest March–May 2023). There is no multi-year training period.
Split	*By time*, never by district or at random. Blocks by target-window date: train 1 Jan 2008–31 Dec 2019, validation 1 Jan–30 Jun 2020, test 1 Jul–31 Dec 2020, with a per-horizon embargo. Main evaluation: rolling-origin forecasts over 2019–2020 (Section 5.6.5).	*By district*, never by season or at random. Whole districts go to training, validation or test (about 70/15/15 of the fields); leave-one-district-out cross-validation if labels are few. "Season" is only a peer-group key, because the survey has a single season.
Feature count	Input per forecast: 12 steps × 6 channels (five anomaly channels and one observed-mask channel) × 28 × 24 cells. Tabular baselines use, per cell, 8 features (B1), 63 (B2) or 73 (B2b).	120 features per field before filtering = 15 variables (10 bands and 5 indices) × 8 temporal summaries (Table 5.14); missingness indicators are extra and not counted in the 120.

in WGS84. A static region mask excludes water and cells outside the study ecoregion from the loss. Records are selected by coordinates (Table 5.2), so the box also covers part of Indian Rajasthan and Indian records inside it are labelled like Pakistani ones: of the 10,364 positive cell-date rows retrieved (Section 5.5.4), 6,986 are Indian and 3,378 Pakistani. Statements about the locust module therefore refer to this box, not to Pakistan's territory. The time step is 16 days, matching MODIS vegetation composites.

Crop study region. The wheat cohort is defined by the wheat crop code of the ADB survey (Pakistan-wide, Rabi 2022–23), not by a coordinate box; a field's coordinates only place it in a district and a Sentinel-2 tile. Peer groups are formed by district (Section 5.7.4), and so are the data partitions (Section 5.7.6); no fixed rectangle is imposed. Sentinel-2 imagery is processed in each image's native UTM

projection and field statistics are recorded in WGS84.

5.5 Data Sources, Acquisition Status and Preliminary Audit

This section reports the state of the data on which both modules depend: how the open FAO observation data were reached, which sources have been secured, and what a first audit of the real files shows. It reports the audit and the retrieval of the multi-year FAO records; the baseline results are in Section 5.6.7.

5.5.1 Access Route and Audit Tool

The prototype is a reproducible audit of the FAO observation files. Its construction followed the real access path, which was not straightforward: the Locust Hub site's own dataset search returned no data layers, and the full historical archive is released through a request form (submitted in September 2026 and pending at the time of writing). A separate open catalogue [44] lists the observation data under a CC BY 4.0 licence, and it provided the files audited below: four life-stage extracts (adults, hoppers, bands, swarms) and the "global" default download. The catalogue also publishes the parameterised query that generates these downloads. That query selects records from a master view by *category* (all, or one of the categories) and *period*, with period options of last week, current month, last month, last three months, last six months, last year, or all. The default download corresponds to the last three months. The same endpoint also accepts an SQL query written against the master view, and this returned the full 2004–2026 history (Section 5.5.4), so the Locust Hub request is no longer on the critical path.

The audit script (Appendix C) reads the extracts and the rolling download and produces the statistics and figures of Section 5.5.3. It applies the coordinate box, reports Pakistani rows through the FAO legacy country code (checked against Pakistan's coordinate extent, see below), avoids the unreliable columns identified below, counts records in the study box by life stage and month, and writes every quoted number to a JSON file so that the report can be regenerated. It is the starting point of the FYP-II label builder.

5.5.2 Data Acquisition Status

5.5.3 Preliminary Data Audit

5.5.3.1 Files and Coverage

The four life-stage files cover only the first months of 2020, a frozen snapshot, and the global download is a rolling three-month window. Neither contains the training years, so the multi-year records were retrieved separately (Section 5.5.4).

Table 5.4: Status of each data source at the time of writing

Source	Status
FAO observation extracts (2020 snapshot)	Downloaded and audited (1 Jan–10 May 2020)
FAO global download, rolling window	Downloaded and audited (1 Jul–28 Sep 2026); a second download proved byte-identical to the first
FAO multi-year records (catalogue query endpoint)	Retrieved for the study box, 2007–2026: 10,364 positive and 56,795 survey-confirmed-absence cell-date rows (Table 5.7)
FAO Locust Hub archive request	Submitted September 2026; pending; no longer needed
MODIS MOD13Q1 and ERA5-Land	Extracted with Earth Engine for 2001–2020 (eight CSV exports); anomaly cube built and tested; MODIS unobserved in 1.7% of cell-steps
ESA CCI soil moisture	Not in the Earth Engine catalogue; replaced by ERA5-Land soil moisture
Locust labels, splits and baselines	Built and covered by 52 automated tests; baselines evaluated (Section 5.6.7)
Sentinel-2 L2A	Source verified; extraction not started
ADB ground-truth survey	Public record located; to be downloaded and profiled
UET Peshawar crop dataset	Public record located; supplementary
Wheat yellow-rust surveillance records	Request to the authors to be sent

Table 5.5: FAO files audited

File	Content	Rows	Dates
`adults_2020.csv`	Adult records, all countries (152 columns)	4,427	1 Jan–10 May 2020
`hoppers_2020.csv`	Hopper records (152 columns)	3,629	1 Jan–10 May 2020
`bands_2020.csv`	Hopper-band records (152 columns)	1,852	1 Jan–9 May 2020
`swarms_2020.csv`	Swarm records (152 columns)	2,567	1 Jan–10 May 2020
`result.csv`	"Global" default download (9 columns)	4,489	1 Jul–28 Sep 2026

5.5.3.2 Pakistani Records in the Study Box

In this audit, Pakistani rows are counted through the FAO legacy code `PA`, after checking that the rows carrying it span 24.4–32.6°N and 61.8–73.8°E, which is Pakistan's extent; the ISO code `PK` returns no rows. This is a counting convenience for the audit only: the locust pipeline itself selects records by coordinates and never by country (Table 5.2). Table 5.6 counts the Pakistani records and those inside the study box.

The 887 in-box records lie on 729 distinct coordinates, and all are flagged with exact location reliability. They are real point observations: even the 326 records whose location name is only "Punjab" sit on 240 distinct coordinates and are not province centroids. Figure 5.2 shows them over the study box and the

Table 5.6: Pakistani records by life stage, 1 Jan–10 May 2020

Life stage	Pakistan (all)	Inside study box
Adults	802	429
Hoppers	1,115	438
Bands	93	19
Swarms	1	1
Total	**2,011**	**887**

0.25° grid; Figure 5.3 shows the monthly counts. Adult records peak in January (273 in the box) and hopper records in April (343 hoppers, 101 adults and 17 bands), the spring breeding that the 2019–2020 test period is meant to capture.

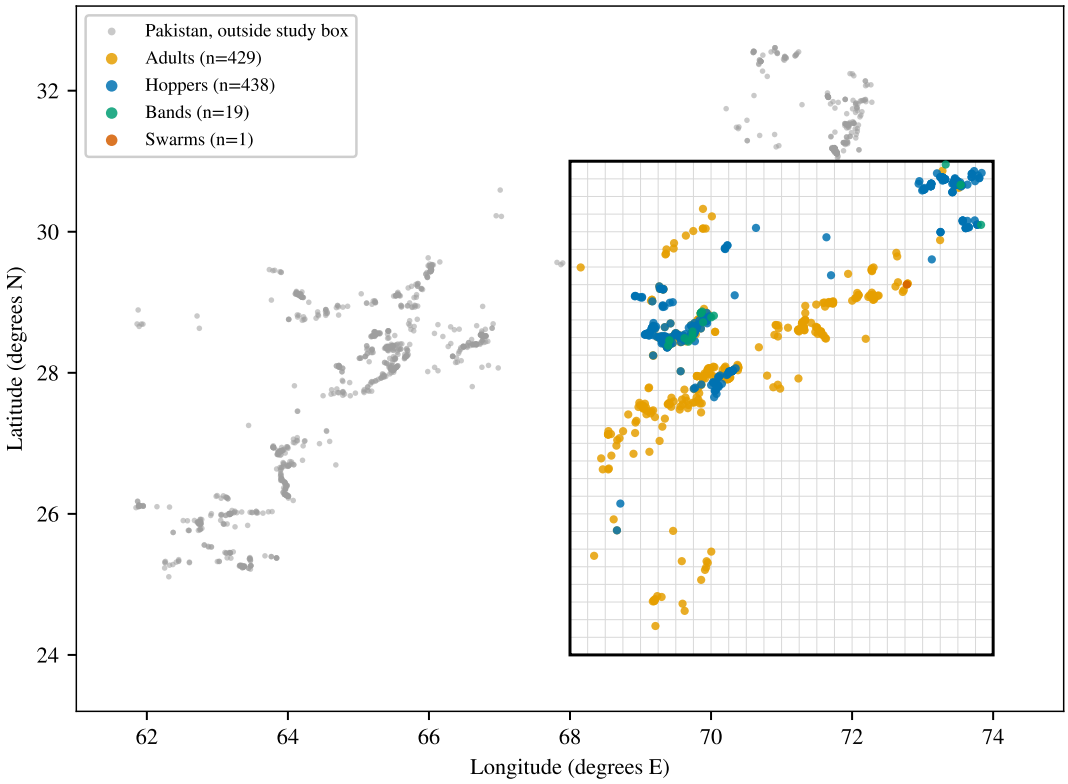


Figure 5.2: FAO records for Pakistan (1 Jan–10 May 2020) over the study box and 0.25° grid. Records inside the box are coloured by life stage; grey points lie outside the box.

5.5.3.3 Data-Quality Findings

1. **Misleading country column.** The column labelled "ISO2 code" holds FAO legacy codes (Pakistan is `PA`). Filtering on `PK` would wrongly conclude that no Pakistani data exist. The label builder therefore selects by coordinates and uses the country field only to report counts.

2. **Misleading date columns.** One start-date column holds the download timestamp (4 October

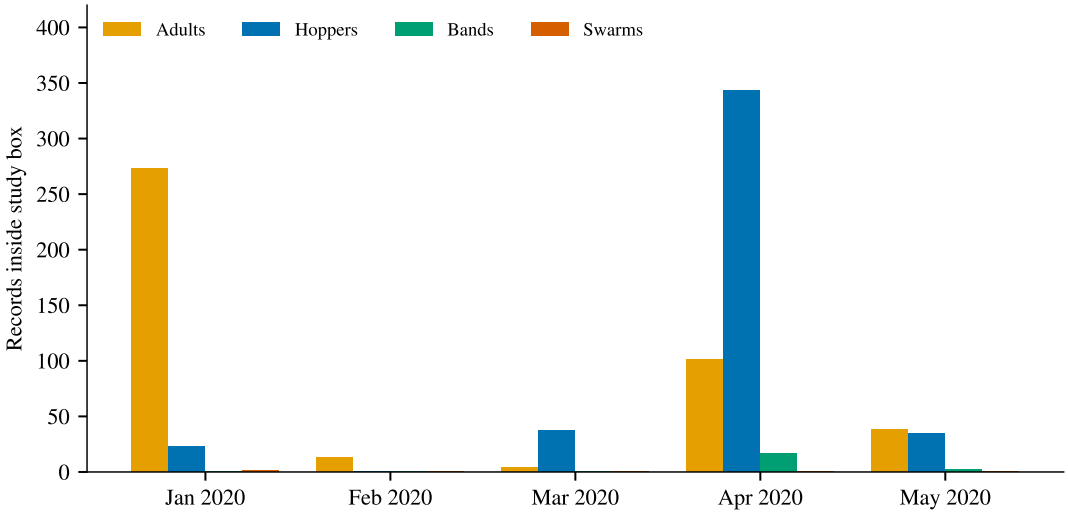


Figure 5.3: Monthly record counts per life stage inside the study box, January–May 2020

2026), not an observation date; the observation-date field is used.

3. **Presence-only data.** The presence flag is 1 for all but five rows across the four files, so the extracts contain essentially no absences.

4. **Unreliable breeding flag.** The `BREEDING` flag equals 1 for only five in-box records (all adults) and never for hopper or band records, which by biology are locally hatched. It is not used as a label; life stage is.

5. **Clustered records.** The 887 in-box records come from only 21 distinct report identifiers, so records are strongly clustered in time and space; the effective number of independent observations is much smaller than 887, which is one reason splits are by time block.

6. **Sparse descriptors.** In the in-box Pakistani records, 103 of the 152 columns are empty or zero in at least 90% of rows; the project uses a handful of fields (Appendix A).

5.5.3.4 Survey-Confirmed Absences in the Rolling Window

The global download contains three categories: `NO LOCUST` (4,157 rows), `ADULT` (317) and `HOPPER` (15). All 470 Pakistani rows in it, every one inside the study box, are `NO LOCUST`. These are survey-confirmed absences—places where surveyors looked and found no locusts—which are a more credible negative class than pseudo-absences. They exist for a period without breeding, so they cannot be used to evaluate 2019–2020, but they show that the global product carries absence information. The multi-year retrieval (Section 5.5.4) shows that the same is true of every year from 2007 on.

5.5.4 Multi-Year Retrieval of the FAO Records

The catalogue's endpoint answers SQL queries over the master view of the observation table, whose columns include category, latitude, longitude, start date, country name, location name and area treated. A single response is capped at 50,000 rows, which truncated the first regional pull at December 2019; the retrieval was therefore split by period and aggregated in SQL to one row per 0.25° cell and observation date (and per category for the positives). One record labelled as Chile but located in the region was dropped as an evident error. Table 5.7 summarises what lies inside the study box (locust module, selected by coordinates).

Table 5.7: FAO records inside the study box by year (cell-date rows: one row per 0.25° cell and observation date)

Year	Survey-confirmed absences	Positives, any life stage	Hoppers or bands
2007 (from Jul)	5	5	2
2008	2,054	5	0
2009	2,074	13	0
2010	2,830	450	119
2011	2,999	110	6
2012	2,395	82	1
2013	2,590	100	0
2014	2,159	44	0
2015	1,809	660	0
2016	2,217	258	1
2017	2,191	31	1
2018	2,189	13	0
2019	1,527	5,525	2,722
2020	15,410	2,984	876
2021–2026	14,346	84	9
Total	**56,795**	**10,364**	**3,737**

Three facts shape the design. First, survey-confirmed absences exist in every year, so negatives need not be invented (Section 5.6.3); the count jumps in 2020, when survey effort rose. Second, positives are rare before the upsurge: from 2008 to 2018 only 1,291 cells-by-16-day-steps hold a positive at any life stage, against 19,736 that hold a survey-confirmed absence, and only 63 hold a hopper or a band, mostly in 2010. Third, positives end in July 2025 while absences continue to September 2026. The positive records come from 49,559 survey records and the absences from 115,963.

5.6 Locust Risk Module

5.6.1 Data Sources and Input Channels

Five anomaly channels and one observed-mask channel, six channels in all, are built for every cell and 16-day step (Table 5.8). The five are expressed as *anomalies* relative to climatology, so the network sees departures from normal and not the strong seasonal cycle.

Table 5.8: Input channels of the locust module

Channel	Source	Native resolution	Reduction to the 0.25°, 16-day cell
NDVI	MODIS MOD13Q1 [20]	250 m, 16-day	Mean of valid pixels in the cell (valid: quality flag good or marginal)
Water-sensitive index	MOD13Q1 NIR (band 2) and mid-infrared (band 7) reflectance (after [47])	250 m, 16-day	Normalised difference of the two bands, then mean over valid pixels
Soil moisture	ERA5-Land volumetric soil water, layer 1 [18]	about 0.1°, daily	Mean over the 16 days
Air temperature	ERA5-Land 2 m temperature [18]	about 0.1°, daily	Mean over the 16 days
Rainfall	ERA5-Land total precipitation [18]	about 0.1°, daily	16-day total, then $\log(1+x)$ to reduce skew
Observed mask	MOD13Q1 quality flags	—	1 if at least 20% of the cell's pixels are valid, otherwise 0

Soil moisture, vegetation and rainfall are the established drivers of breeding [7, 8, 23]; temperature governs development speed. Reduction to the cell grid is done server-side in Google Earth Engine [45] where possible, so that only small arrays are exported. The coverage of each product was verified before export (risk R5 in Section 4.10): ESA CCI soil moisture is not available in Earth Engine [48, 49] and the ERA5 daily collection there ends on 9 July 2020 [50], so the three weather and soil channels come from ERA5-Land. Eight exports (MODIS and ERA5-Land, each in four five-year chunks covering 2001–2020) give 672 cells $\times$ 23 steps per year; MODIS is unobserved in 1.7% of cell-steps.

Anomaly computation. For channel c, cell i, and 16-day step-of-year k, the anomaly of an observation x is

$$z_c(i,t) = \frac{x_c(i,t) - \mu_c(i,k)}{\sigma_c(i,k) + \varepsilon}, \qquad k = \kappa(t), \tag{5.1}$$

where μ_c and σ_c are the mean and standard deviation of x_c at the same cell and step-of-year over the *training years only* (2001–2019), and ε is a small constant; anomalies are clipped to ± 5. The clima-

tology never uses 2020, so nothing from the evaluation year enters the inputs. It is fitted once and not refitted at every rolling origin, so 2019 *input* data (not labels) contribute to the reference used for 2019 origins; this is a small effect on a 19-year reference and is stated as a limitation (Section 5.9). Missing values (clouds, retrieval gaps) are set to zero anomaly after the mask channel records them, so the network can tell "normal" from "unobserved".

5.6.2 Label Construction

Source records. FAO observation records give latitude, longitude, date and life stage (adults, hoppers, bands, swarms); records of category `NO LOCUST` mark surveyed places where none was found. The audit (Section 5.5.3) established the rules for using them: (i) select records by coordinates (the study box), not by the country-code or country-name column, because the 2020 files use FAO legacy codes in a column labelled "ISO2" (Pakistan is `PA`, not `PK`) and because the box crosses the border; (ii) use the observation date, not the timestamp columns of the 2020 files, which record download time; (iii) drop records without valid coordinates or dates and the one Chile-labelled record (the master view has no location-reliability column, so the exact-location filter applied to the 2020 extracts is not available; all in-box records of those extracts were exact); (iv) do not use the `BREEDING` flag as a label, because it is set for only five in-box records (all adults) and never for hopper or band records.

Target definition. The *primary target* is presence at any life stage: a cell is positive in a target window if at least one adult, hopper, band or swarm record falls in it. Hoppers and hopper bands cannot fly, so a record of either is evidence that eggs were laid and hatched locally; presence of these stages is the *secondary* (breeding-stage) target and is reported separately. The primary target was changed from breeding stage to any stage because the study box holds only 63 hopper or band cell-steps in 2008–2018 (Section 5.5.4), too few to train or evaluate on outside the upsurge. For a forecast issued at origin t_0 (the end of the last input composite) and a horizon $h \in \{28, 42, 56\}$ days, the target window is $W_h = [t_0+h,\, t_0+h+16)$ days. Defining windows by calendar date, not by composite index, keeps the horizons exactly 28, 42 and 56 days and ensures that no input composite overlaps any target window.

Three-valued labels. Each cell, origin and horizon has a label $y_h \in \{0,1\}$ and a mask $m_h \in \{0,1\}$, where $m_h = 0$ means *unknown* and is excluded from the loss and the metrics. A cell is positive ($y = 1, m = 1$) if it contains a qualifying record in the window; negative ($y = 0, m = 1$) only if it is a designated negative (below); and every other cell is unknown. Algorithm 1 gives the procedure and Figure 5.4 illustrates it. In the training block about 12% of cell-origin pairs are known, and about 12% of the known cells are positive for the any-stage target; the corresponding positive shares are about 45% in the validation block (January–June 2020, 28-day horizon) and about 12% in the test block (July–December 2020).

Input: FAO records R; target stages S (any stage, or hopper and band); grid G; horizons $\mathscr{H} = \{28, 42, 56\}$; absence set A (survey-confirmed or pseudo)
Output: label tensors Y_h and masks M_h for every forecast origin t_0
Filter R to the study box (by coordinates) and valid observation dates;
foreach *origin* t_0 *(every 16 days)* **do**
  foreach $h \in \mathscr{H}$ **do**
    $W \leftarrow [t_0 + h,\ t_0 + h + 16)$ days; $Y_h \leftarrow 0$; $M_h \leftarrow 0$;
    foreach *record* $r \in R$ *with stage* $\in S$ *and date* $\in W$ **do**
      $i \leftarrow \text{cell}(r)$; $Y_h[i] \leftarrow 1$; $M_h[i] \leftarrow 1$;
    end
    foreach *absence* $a \in A$ *with date* $\in W$ *and* $Y_h[\text{cell}(a)] = 0$ **do**
      $M_h[\text{cell}(a)] \leftarrow 1$;
    end
    set $M_h \leftarrow 0$ outside the region mask;
  end
end

Algorithm 1: Construction of masked locust label tensors

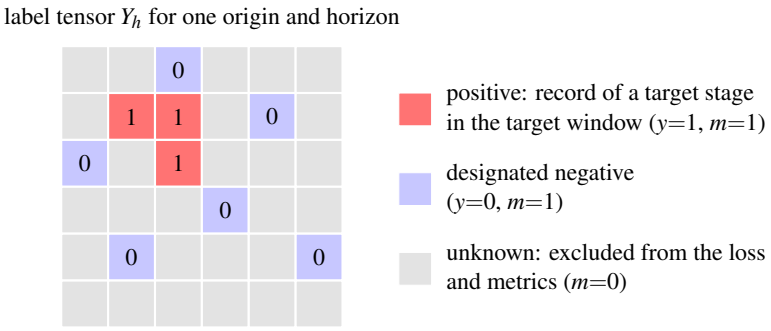


Figure 5.4: Three-valued labels: only known cells contribute to training and evaluation; most cells of the grid are unknown

5.6.3 Negative Examples and the Presence-Only Problem

FAO records say where locusts were seen, never where they were absent, so negatives have to be constructed, and the choice affects the result [24, 22]. Three strategies are compared as an ablation (RQ2):

N1. Survey-confirmed absences. The global FAO product contains "NO LOCUST" records: surveyed places where locusts were looked for and not found. In the first audit these made up 4,157 of 4,489 rows of the rolling three-month window, including all 470 Pakistani rows. The multi-year retrieval contains 56,795 cell-date rows (115,963 survey records) inside the box from 2007 to 2026 (Table 5.7), so these are the primary negatives. They still reflect where surveyors went.

N2. Environmentally stratified pseudo-absences. Cells in the same region, step-of-year and year as a positive that have no record of the target stage in a surrounding window, so the model must separate breeding from non-breeding *within* the desert margin rather than desert from non-desert.

N3. Random in-region pseudo-absences (control). Uniform sampling from the unlabelled cells, included to show how much the stratification matters.

In the primary configuration only N1 provides negatives and unknown cells remain masked. N2 and N3 are planned ablations and have not yet been run; all results in this report use N1.

5.6.4 Models

Baseline ladder. Models are built in order of complexity, each answering whether the next step is needed (Table 5.9). B0, B1, B2, B2b, P, S2 and S2b are implemented and evaluated (Section 5.6.7); B3 and M are specified but not yet implemented. No coordinates are used as features, so B0 is the only model with a static spatial prior.

Table 5.9: Locust model ladder

ID	Model	Description	Uses space/time?
B0	Climatological prior	Probability of a positive record per cell and calendar month of the target window, estimated from training labels and smoothed over 3×3 cells	Neither
B1	Logistic regression	Current-step anomalies of the five channels, the mask and a month encoding (8 features per cell)	Neither
B2	Random forest [51]	Flattened last 12 steps of the five channels, the mask and a month encoding (63 features per cell), in the style of LocustLens [11]	Time, no neighbours
B2b	LightGBM, gradient-boosted trees	B2 features plus 3×3 and 7×7 neighbourhood means of the current anomalies (73 features)	Time and neighbours
B3	Per-cell LSTM [25]	Sequence of 12 steps for one cell	Time, no neighbours
P	Persistence (recent sightings)	No fitted parameters: the cell's own record history over the 12 steps, halved at each older 16-day step, plus a smaller share from its 3×3 neighbourhood	Past records only
S2, S2b	Random forest and LightGBM with sightings	B2 and B2b features plus 12 steps of sightings (any stage, hopper or band, surveyed) and nearby-record flags (111 features)	Time and neighbours, with past records
M	ConvLSTM [17]	Whole-region sequence, three horizon heads	Both

ConvLSTM architecture. The input is a sequence $X \in \mathbb{R}^{T \times C \times H \times W}$ with $T = 12$ steps (about six months), $C = 6$ channels (the five anomaly channels and the observed-mask channel) and $H \times W = 28 \times 24$. A ConvLSTM layer keeps a cell state $\mathscr{C}_t$ and hidden state $\mathscr{H}_t$ that are spatial maps, updated by

convolutions $*$:

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * \mathscr{H}_{t-1} + b_i), \qquad f_t = \sigma(W_{xf} * X_t + W_{hf} * \mathscr{H}_{t-1} + b_f), \tag{5.2}$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * \mathscr{H}_{t-1} + b_o), \qquad g_t = \tanh(W_{xg} * X_t + W_{hg} * \mathscr{H}_{t-1} + b_g), \tag{5.3}$$

$$\mathscr{C}_t = f_t \circ \mathscr{C}_{t-1} + i_t \circ g_t, \qquad \mathscr{H}_t = o_t \circ \tanh(\mathscr{C}_t), \tag{5.4}$$

where $\circ$ is the elementwise product. (This is the common variant without the peephole terms of the original paper [17].) Two layers with 32 and 64 hidden channels and 3×3 kernels encode the sequence; the final hidden state of the second layer, $\mathscr{H}_T \in \mathbb{R}^{64\times28\times24}$, feeds three 1×1 convolutions with sigmoid outputs, one per horizon, each producing a risk map $P_h \in [0,1]^{28\times24}$. By construction the model has $43{,}904 + 221{,}440 + 195 = 265{,}539$ parameters ($\approx 2.7 \times 10^5$), small enough to train on a laptop or a free-tier GPU. All horizons share the encoder, so differences between horizons reflect the target, not different features. Figure 5.5 summarises the architecture.

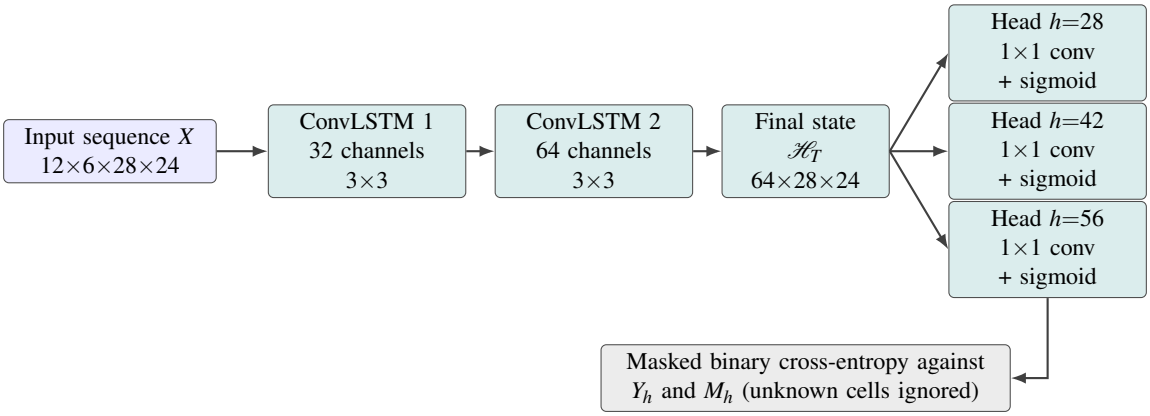


Figure 5.5: ConvLSTM forecaster: a shared encoder and three horizon heads

Loss. With predictions $p_h(i)$, labels $y_h(i)$ and masks $m_h(i)$, the training loss is a masked, positive-weighted binary cross-entropy:

$$\mathscr{L} = -\frac{\sum_h \sum_i m_h(i) \left[w_+ \, y_h(i) \log p_h(i) + (1 - y_h(i)) \log(1 - p_h(i)) \right]}{\sum_h \sum_i m_h(i)}, \tag{5.5}$$

where w_+ is the ratio of negative to positive known cells in the training set.

Training configuration. Table 5.10 lists the initial settings. They are starting points, adjusted only on the validation partition.

A caution on sample size. Forecast origins occur every 16 days, giving only about 275 training origins per horizon for 2008–2019 (and 45 rolling-evaluation origins), and consecutive origins are highly correlated. This is small for a deep model. The design responds with a small network, early stopping, spatial

Table 5.10: Initial training configuration of the ConvLSTM

Setting	Value
Optimiser	Adam, learning rate 10^{-3}, weight decay 10^{-4}
Batch and epochs	Batch of 8 sequences; at most 100 epochs
Early stopping	Patience 10 epochs on PR-AUC averaged over horizons, measured on the validation block (fixed blocks) or on the latest labelled months before the cut-off (rolling-origin refits)
Regularisation	Dropout 0.1 between layers; random 16×16 spatial crops during training only
Class imbalance	Positive weight w_+ as in Eq. (5.5); no synthetic oversampling
Seeds	Five seeds; mean and standard deviation reported

crops, strong baselines (B2, B3 and the persistence rule) and a stated rule: if the ConvLSTM does not beat them, that result will be reported as it stands.

5.6.5 Data Splits and Leakage Control

In the locust module the data are split *by time*, never by district and never at random, because adjacent steps and neighbouring cells are nearly identical [52] (Figure 5.6).

Fixed development blocks. A forecast sample (origin, horizon) belongs to a block by the dates of its *target window*: training from 1 January 2008 to 31 December 2019, validation from 1 January to 30 June 2020, and test from 1 July to 31 December 2020. Training starts in 2008 because the FAO records in the box begin in August 2007; the inputs and the climatology reach back to 2001. A sample whose 16-day target window straddles a block boundary belongs to no block (the *embargo*). Because the window starts 28, 42 or 56 days after the origin, the embargo is applied per origin and horizon and removes at most the 72 days before each boundary. The blocks hold 275, 10 and 11 origins per horizon.

Rolling-origin evaluation (main evaluation). Ten and eleven origins per horizon are a thin basis, and most of the 2019–2020 upsurge falls in the training block. The main evaluation is therefore rolling-origin: the 45 origins per horizon whose target windows lie in 1 January 2019–31 December 2020 are scored in time order; every model is refitted every six origins using only samples whose target windows had ended by the first origin of that group, and each origin is scored exactly once (Section 5.6.7). This refit schedule is conservative, because later origins in a group do not receive newer labels. Hyperparameters are fixed in advance and are not tuned on the rolling results. Because the rolling period overlaps the validation and test blocks, 2020 no longer provides an untouched test set: the fixed blocks serve for development and the rolling-origin results are the reported evaluation.

Leakage controls. The climatology and every learned parameter are fitted without 2020 inputs or labels. Past locust records are *not* used as inputs in the primary models, since a model fed last season's sightings

could simply copy them; the recent-sightings models are run separately and labelled as such. Records dated on or after the forecast origin are excluded from the sightings inputs by construction, but FAO records can reach the database late, so the sightings baselines assume more information at forecast time than an operational system would have.

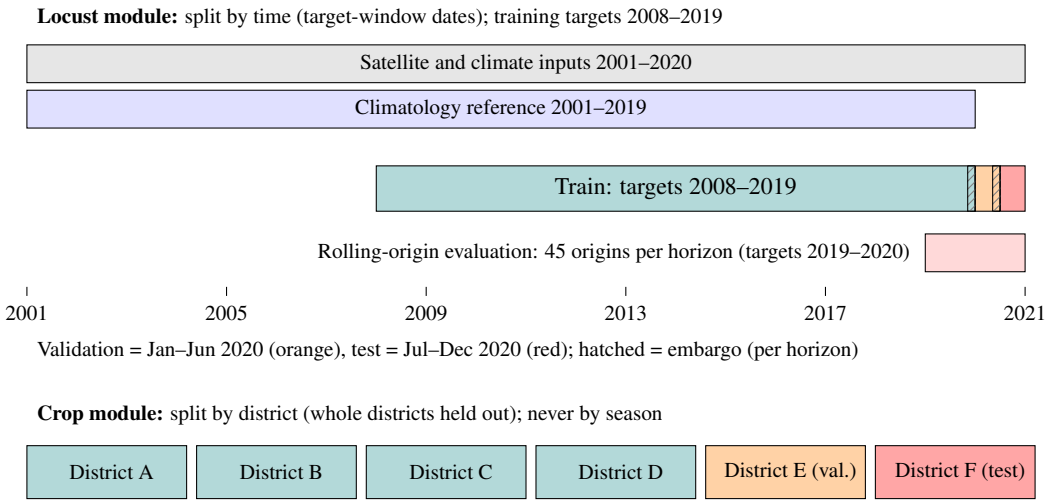


Figure 5.6: Leakage-controlled partitions. Locust module: time blocks with a per-horizon embargo (hatched), plus rolling-origin forecasts over 2019–2020. Crop module: whole districts held out; never a split by season

Data-sufficiency check (done). Locust activity is episodic and the years before the 2019–2020 upsurge were comparatively quiet. After retrieval, positives were counted (Section 5.5.4): from 2008 to 2018 the box holds 1,291 positive cell-steps at any life stage but only 63 for hoppers or bands. This led to the any-stage primary target and to the rolling-origin evaluation (Table 5.3). Training on the wider Southwest Asian recession area (Pakistan together with neighbouring countries in the FAO data) remains a fallback for the breeding-stage target (Section 4.10).

5.6.6 Evaluation

Positives are rare, so accuracy and ROC-AUC would flatter every model [21]. Metrics are computed on known cells only, separately for each horizon:

- **PR-AUC** (average precision), the primary threshold-free measure.
- **F1** at the threshold that maximises F1 on the validation set, then frozen.
- **Recall at a fixed alert area:** the fraction of positive cells captured when only the top $q\%$ of cells by predicted risk are flagged, for $q \in \{5, 10, 20\}$, which expresses operational value.
- **Spatially tolerant versions** of the above, in which a prediction counts as a hit if a positive lies in the same or an adjacent cell, following the buffered scoring of [7].

- **Calibration:** Brier score and reliability curves.
- **Event lead time (planned, not yet computed):** for each evaluation-period event, the number of days between the earliest forecast origin at which predicted risk stayed above the threshold and the first record.

Every PR-AUC is read against the *prevalence* of positives among known cells, which is the score of a random ranking; *lift* is PR-AUC divided by prevalence, and a lift of 1 means no skill. Uncertainty is estimated by a block bootstrap that resamples whole forecast origins (500 resamples); with about two years of evaluation data the intervals are wide and are reported as such. **Success criterion.** The ConvLSTM is considered to show an advantage only if its rolling-origin PR-AUC exceeds those of the persistence baseline and of the better of B2 and B3 by more than the bootstrap interval for at least two of the three horizons. Otherwise the report states that no advantage was demonstrated. Persistence was added to the criterion after the baseline results below were seen and before any ConvLSTM result existed. Figures from [10] (F1 = 81.53%) are quoted as a reference point only, never as a head-to-head comparison, because they come from African data and a different label construction.

5.6.7 Preliminary Baseline Results (Locust Module)

Protocol. The baselines of Table 5.9 were evaluated by the rolling-origin protocol of Section 5.6.5: 45 forecast origins per horizon with targets in 2019–2020, each model refitted every six origins on earlier labels only, survey-confirmed absences as negatives (N1), and metrics on known cells. To limit run time the random forests use 100 trees in these runs (300 in the fixed-block development runs), each model has one seed, and no setting was tuned on these results. The per-cell LSTM and the ConvLSTM have not been run, so nothing below concerns them.

Any-stage target (primary). Table 5.11 and Figure 5.7 give the results. The environment-only models B0–B2b score *below* prevalence at every horizon (PR-AUC 0.27–0.32 against 0.37, a lift of 0.73–0.86), and their recall at a 10% alert area (0.03–0.07) is below the 0.10 that a random ranking would give. Of the models with sightings, only the parameter-free persistence rule (P) is above prevalence: its PR-AUC is 0.53, 0.47 and 0.43 at 28, 42 and 56 days (lift 1.42, 1.28, 1.17), with 17% of positives caught in a 10% alert area at 28 days and 35% when a neighbouring cell counts (0.14 and 0.32 at 56 days). Its lead shrinks with the horizon and, because the intervals are wide, its interval clears prevalence only at 28 days. Adding the environment to the sightings (S2, S2b) gives 0.40–0.42 at 28 days and 0.31–0.34 at 56 days, better than the environment alone but worse than persistence: a learned model on about 290 correlated training origins does not improve on the simple rule.

Breeding-stage target (secondary). The pattern is the same (Table 5.12). Environment-only models

Table 5.11: Any-stage target (primary): rolling-origin PR-AUC (95% block-bootstrap interval); bold if the whole interval lies above prevalence

Model	28 days	42 days	56 days
Prevalence (random ranking)	0.371	0.367	0.368
B0 climatological prior	0.317 (0.22–0.47)	0.317 (0.22–0.47)	0.317 (0.22–0.47)
B1 logistic regression	0.303 (0.20–0.45)	0.293 (0.20–0.43)	0.299 (0.20–0.44)
B2 random forest	0.280 (0.19–0.40)	0.276 (0.19–0.40)	0.269 (0.18–0.39)
B2b LightGBM + neighbours	0.290 (0.20–0.43)	0.279 (0.19–0.41)	0.278 (0.19–0.40)
P persistence (sightings only)	**0.526 (0.39–0.68)**	0.470 (0.33–0.63)	0.430 (0.30–0.59)
S2 random forest + sightings	0.403 (0.28–0.57)	0.342 (0.23–0.50)	0.314 (0.21–0.45)
S2b LightGBM + sightings	0.418 (0.29–0.59)	0.344 (0.23–0.51)	0.338 (0.23–0.49)

score 0.11–0.15 against a prevalence of 0.17 (lift 0.64–0.84), the sightings-plus-environment models are at prevalence or below, and persistence is the only model above it: 0.37, 0.29 and 0.24 at 28, 42 and 56 days (lift 2.1, 1.7, 1.4), with 29% of positives in a 10% alert area at 28 days. Only at 28 days does its interval clear prevalence. Hopper and band records are rare and cluster in the upsurge, so these figures rest on very few independent events.

Table 5.12: Breeding-stage target (secondary): rolling-origin PR-AUC (95% block-bootstrap interval); bold if the whole interval lies above prevalence

Model	28 days	42 days	56 days
Prevalence (random ranking)	0.174	0.173	0.172
B0 climatological prior	0.134 (0.08–0.23)	0.133 (0.08–0.22)	0.134 (0.08–0.22)
B1 logistic regression	0.147 (0.09–0.24)	0.142 (0.09–0.24)	0.139 (0.08–0.23)
B2 random forest	0.123 (0.08–0.19)	0.124 (0.08–0.19)	0.121 (0.07–0.18)
B2b LightGBM + neighbours	0.118 (0.07–0.18)	0.111 (0.07–0.17)	0.111 (0.07–0.17)
P persistence (sightings only)	**0.368 (0.20–0.55)**	0.290 (0.15–0.47)	0.240 (0.12–0.40)
S2 random forest + sightings	0.169 (0.10–0.29)	0.148 (0.09–0.26)	0.134 (0.08–0.22)
S2b LightGBM + sightings	0.178 (0.10–0.29)	0.129 (0.08–0.20)	0.122 (0.07–0.19)

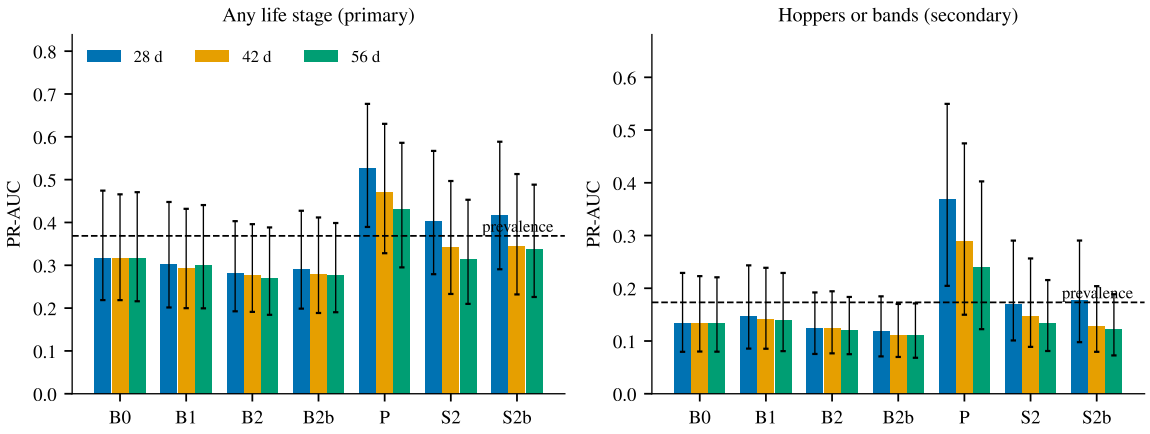


Figure 5.7: Rolling-origin PR-AUC of the locust baselines (bars: 28, 42 and 56 days; whiskers: 95% block-bootstrap interval). The dashed line is prevalence, the score of a random ranking. B0–B2b use the environment only; P uses sightings only; S2 and S2b use both

The pooled score hides large period effects. Table 5.13 splits the 28-day any-stage result by half-year. In the second half of 2019, the height of the upsurge, 87% of known cells are positive and every model scores about 0.9, which says little; in the second half of 2020 only 3% are positive and no model reaches 0.13. In between, no model leads consistently: the sightings-plus-environment model is best in the first half of 2019 and logistic regression in the first half of 2020. The ranking of Table 5.11 is therefore a statement about the pooled period and not a uniform one.

Table 5.13: Any-stage target, 28-day horizon: PR-AUC by half-year of the rolling-origin period

Model	2019 H1	2019 H2	2020 H1	2020 H2
Origins	12	11	12	9
Prevalence	0.334	0.869	0.411	0.033
B0 climatological prior	0.506	0.911	0.402	0.038
B1 logistic regression	0.607	0.908	0.500	0.079
B2b LightGBM + neighbours	0.561	0.885	0.419	0.063
P persistence (sightings only)	0.525	0.918	0.436	0.089
S2b LightGBM + sightings	0.747	0.917	0.420	0.120

Reading the results. (i) The result agrees with the fixed development blocks: on the validation block (January–June 2020, 10 origins per horizon) every model scored between 0.40 and 0.55 against a prevalence of 0.44–0.46 for the any-stage target. (ii) The environmental anomalies carry little usable signal in this data: in the training block the best single channel, the rainfall anomaly, had a ROC-AUC of 0.62 and the NDVI anomaly 0.50, and a gradient-boosted model fitted to the training block reproduced it almost perfectly (PR-AUC 0.97) while scoring at chance on later years, which indicates overfitting to years that do not resemble the evaluation years. (iii) Recent sightings help mainly because the upsurge persisted for months, not because the models understand breeding conditions; in the validation block, cells with a record in the previous three steps were no more likely to be positive (42%) than other known cells (47%). (iv) Known cells are where surveyors went, and survey effort and control operations changed between years. These are preliminary baseline results on one event; they do not show what a ConvLSTM can do. The ConvLSTM will be judged against persistence, and if it cannot beat that rule the report will say so (Section 5.6.6).

5.7 Crop Abnormality Screening Module

5.7.1 Task, Data Layers and Evidence Tiers

The module asks whether a wheat field appears *normal* or *spectrally abnormal* relative to appropriate peers. It does not name a disease, because disease, drought, nutrient deficiency, waterlogging, late sowing and normal senescence overlap spectrally [13]. The data are deliberately separated into two layers:

- **Dataset A (screening cohort).** Wheat identity, location, date and growth stage from public Pakistani crop-ground-truth data, plus the Sentinel-2 time series. It is unlabelled for health and is sufficient to fit the model and produce a continuous anomaly score.
- **Dataset B (validation subset).** Independently assessed fields with a date-matched Normal or Abnormal condition, from disease surveillance or a targeted expert inspection. It is required to choose a binary threshold and to report classification metrics.

Normal means a field assessment found no material crop-health abnormality at the matched time; *Abnormal* means substantial disease or another significant stress was verified. A spectral anomaly is never accepted as its own ground truth. Claims are tied to three tiers of evidence: (1) **anomaly-ranking feasibility**, supported when atypical trajectories are ranked consistently; (2) **binary Pakistan screening performance**, supported only by independently labelled Pakistani test fields; and (3) **operational generalisation**, supported only after evaluation on unseen districts and at least one further growing season. If no local labels can be obtained, only tier 1 and expert-reviewed examples are claimed.

5.7.2 Cohort Construction

Sources. Wheat records are selected from the ADB ground-truth survey using its wheat crop code [33]; the UET Peshawar crop dataset [34] provides a supplementary local wheat mask and a test bed for the extraction code. Sentinel-2 Level-2A surface reflectance is taken from the harmonised Copernicus collection (for example `COPERNICUS/S2_SR_HARMONIZED` in Earth Engine) [9, 45].

Observation unit. The preferred unit is a wheat-field polygon in one season, eroded inwards by at least one 10 m pixel before statistics are taken, which limits contamination from roads, canals and neighbouring crops [4]. Where only a survey point exists, a 3×3 window of 10 m pixels around the point is used and the record is flagged *point-derived* and reported separately. Every field–season pair has one identifier. Partitions are made by district (Section 5.7.6), so all fields and dates of a district stay together.

Inclusion and exclusion. A sample is included if it is labelled wheat in the Pakistani source, has valid coordinates or a usable polygon and an interpretable date, lies in the local wheat season with at least 60% usable temporal coverage, retains enough valid pixels after cloud masking, and is not dominated by mixed crops, built surfaces, open water or geolocation error. Exclusions and their reasons are stored, not silently dropped; duplicate coordinates, impossible dates, implausible reflectances and coordinate-system errors are checked before extraction. The wheat season runs approximately from sowing in October–November to harvest in March–May and is fixed per district from the growth-stage records.

5.7.3 Preprocessing

1. **Masking.** The Scene Classification Layer and the cloud-probability layer remove cloud shadow, medium- and high-probability cloud, cirrus and snow or ice. Reflectances are converted to unit values with the product scale factor.

2. **Band alignment.** Bands B2, B3, B4 and B8 are kept at 10 m; B5, B6, B7, B8A, B11 and B12 are resampled from 20 m to the 10 m grid by bilinear interpolation. This harmonises the feature stack but creates no new spatial information.

3. **Compositing.** Valid observations are aggregated into 10-day median composites over the wheat season.

4. **Gaps.** A gap of at most two composites is filled by linear interpolation. Longer gaps are filled with the median of the matching peer group and flagged by a binary missingness feature. Samples with less than 60% valid coverage are excluded.

5. **No augmentation.** No colour, geometric or random-image augmentation is applied, since it would alter physically meaningful reflectances.

5.7.4 Indices, Features and Peer Groups

Five indices are derived from the harmonised bands; NDVI [53] is kept as a baseline and not as the sole indicator:

$$\text{NDVI} = \frac{B8 - B4}{B8 + B4}, \qquad \text{NDRE} = \frac{B8A - B5}{B8A + B5}\ [54], \tag{5.6}$$

$$\text{GNDVI} = \frac{B8 - B3}{B8 + B3}\ [55], \qquad \text{NDMI} = \frac{B8A - B11}{B8A + B11}\ [47], \tag{5.7}$$

$$\text{OSAVI} = \frac{1.16\,(B8 - B4)}{B8 + B4 + 0.16}\ [56]. \tag{5.8}$$

NDRE uses the red-edge band, which saturates less than NDVI in dense canopies, and NDMI adds moisture sensitivity; none of the indices is specific to disease.

Features and how they are counted (crop module). The 15 variables are the 10 spectral bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12) and the 5 indices above. For each variable and each 10-day composite, the field median (typical behaviour) and the within-field interquartile range (heterogeneity, which was the most frequently detected anomaly category in the precedent study [4]) are computed. Eight temporal summaries are then taken per variable (Table 5.14), giving $15 \times 8 = 120$ features per field before filtering. Missingness indicators for gap-filled composites are stored separately and are not

part of the 120. Features with $|r| > 0.95$ are dropped on the training partition (Table 5.15), so the number finally used is smaller and will be reported with the results.

Table 5.14: The eight temporal summaries per variable (crop module): $15 \times 8 = 120$ features

#	Summary	Computed from
1	Seasonal median	Field-median series
2	Seasonal median of the within-field IQR (heterogeneity)	Within-field IQR series
3	Maximum	Field-median series
4	Date of the maximum	Field-median series
5	Area under the curve	Field-median series
6	Largest short-term rise (composite to composite)	Field-median series
7	Largest short-term fall (composite to composite)	Field-median series
8	Slope over the growth stage assigned to the field (one slope per variable)	Field-median series

Peer groups. Normal development differs across districts, seasons and sowing dates, so fields are compared only with peers of the same district (or agro-ecological zone), season and crop stage [30]; because the survey has a single season, the season key is constant here. Each feature f_j of field a in peer group g is converted to a robust score

$$\tilde{f}_j(a) = \frac{f_j(a) - \text{median}_g(f_j)}{\text{IQR}_g(f_j) + \varepsilon}, \tag{5.9}$$

with the group statistics computed on the training partition only. Groups that are too small are merged with the next coarser level; the minimum size is fixed empirically in FYP-II. This responds to the stated weakness of anomaly detectors: if abnormality is common (yellow rust affected 83% of surveyed fields in one region [3]), only a within-peer comparison and external calibration can separate it from the norm.

5.7.5 Isolation Forest and Threshold Calibration

An Isolation Forest builds trees by choosing a random feature and a random split value; points that are isolated after few splits are anomalous [19, 27]. For a sample x and subsample size n, the anomaly score is

$$s(x,n) = 2^{-E[h(x)]/c(n)}, \qquad c(n) = 2H(n-1) - \frac{2(n-1)}{n}, \tag{5.10}$$

where $h(x)$ is the path length, $H(i) \approx \ln i + 0.5772$ is the harmonic number, and scores near 1 mean "anomalous". The implementation is scikit-learn's `IsolationForest` [46], whose score is sign-flipped so that larger means more abnormal. Table 5.15 gives the configuration. The primary configuration fits one forest on peer-normalised features; an ablation fits one forest per peer group.

Table 5.15: Isolation Forest configuration

Parameter	Setting
Number of trees	500 (sensitivity check at 200 and 1,000)
Subsample per tree	$\min(256, n_{\text{train}})$, as in the original and precedent studies
Features / bootstrap	All retained features; no bootstrap
Contamination	`auto` for fitting; *not* used as the health threshold
Feature filtering	Remove one of any pair with $\lvert r\rvert > 0.95$ (fitted on training data)
Seeds	42, 52, 62, 72, 82; mean and standard deviation reported
Binary threshold τ	Chosen on labelled Pakistani validation fields by maximum F1, with Abnormal recall as a safety-oriented secondary criterion; frozen before testing

A fixed contamination percentage is not assumed because prevalence varies widely between districts and seasons [3]. The raw score is the primary output; the binary class exists only after calibration (Algorithm 2).

Input: wheat cohort $\mathscr{F}$ with geometries; Sentinel-2 L2A; peer-group keys; labelled validation subset $\mathscr{V}$ (optional)
Output: anomaly score s_a for every field; threshold τ if $\mathscr{V}$ exists
Partition $\mathscr{F}$ into train/validation/test by district (all fields of a district go to the same partition);
foreach *field a* **do**
  mask clouds; composite to 10-day medians; align bands; compute indices and zonal median/IQR; derive temporal features x_a;
end
Fit peer statistics on training fields; transform $x_a \to \tilde{x}_a$ (Eq. 5.9);
Drop features with $\lvert r\rvert > 0.95$; fit `IsolationForest` on training $\tilde{x}_a$;
$s_a \leftarrow -\text{score_samples}(\tilde{x}_a)$ for all fields;
if $\mathscr{V} \neq \emptyset$ **then**
  choose τ maximising F1 on validation scores; freeze τ;
end
Evaluate once on the test partition;

Algorithm 2: Crop field scoring and threshold calibration

5.7.6 Baselines and Partitioning

Three leakage-resistant comparisons are planned: (i) **peer-median deviation**, the mean absolute robust score of Eq. (5.9) over all features; (ii) an **NDVI-only rule** that ranks fields by NDVI decline from the seasonal peak, $(\text{NDVI}_{\text{max}} - \text{NDVI}_{\text{late}})/\text{NDVI}_{\text{max}}$, compared with the peer median; and (iii) the **multi-band Isolation Forest**. If at least 200 independently labelled Pakistani fields with both classes are obtained, a supervised random forest (500 trees, balanced class weights, minimum leaf size in $\{1,2,4\}$, depth in $\{\text{none}, 10, 20\}$, chosen by grouped validation only) is added to show whether limited supervision beats unsupervised ranking. A foreign supervised model (for example RustMAE [31]) may be applied zero-shot to the Pakistani test set, but only as a labelled domain-transfer baseline [12].

In the crop module all partitions are made *by district*, before imputation, scaling, interpolation parameters, peer statistics or model fitting: a district goes wholly to training, validation or test, so no field of a

held-out district is seen in fitting. The preferred split is about 70% training, 15% validation and 15% test of the fields, by district. The training population is the unlabelled cohort of Dataset A; validation and test are drawn from the labelled Dataset B. Because the public wheat survey covers one season, a split by season is not possible and none is used; if too few labels exist for the district split, leave-one-district-out grouped cross-validation replaces it.

5.7.7 Evaluation

With a complete binary ground truth on the test set (Abnormal as the positive class), the module reports

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2PR}{P+R}, \quad \text{Specificity} = \frac{TN}{TN+FP}, \tag{5.11}$$

balanced accuracy $= \frac{1}{2}(\text{Recall} + \text{Specificity})$, and PR-AUC as the principal threshold-free measure, with ROC-AUC secondary. Recall matters because missed abnormal fields reduce the value of a screening tool, and precision controls unnecessary inspections. When only a subset of fields can be inspected, accuracy and recall cannot be estimated reliably, and the module reports

$$\text{Precision@}k = \frac{\text{confirmed abnormal fields among the top-}k}{k}, \tag{5.12}$$

on an inspected sample that includes high-, medium- and low-score fields so that it is not restricted to obvious cases. Results are stratified by district, growth stage, sowing window and polygon versus point extraction.

5.7.8 Confounders and Early-Season Scoring

Several non-health causes can raise an anomaly score: late sowing, normal senescence, clouds, mixed pixels, and unusually vigorous but healthy crops. The ADB wheat season (Rabi 2022–23) followed the 2022 floods, which delayed Rabi sowing in flood-affected areas of Sindh [35]; stage-matched peers should absorb much of this, but residual late-sowing effects are analysed explicitly, and districts with documented flood-delayed sowing offer a label-free sanity check (their scores should be higher), subject to confirming that they appear in the survey.

The features above summarise a whole season, so scoring is retrospective. To study whether the module can warn *during* the season, an **as-of-date** mode will compute features and peer statistics from composites up to date t only and measure how early the score separates abnormal from normal fields. The precedent study showed that parcel anomalies can already be detected on part of a season [4]. Because the evaluation needs verified labels, the lead-time comparison with the locust module is reported only if the labels permit it.

5.8 Experimental Plan and Evaluation Protocol

Because neither the ConvLSTM nor the crop module has been trained, this section fixes the remaining experiments, the reporting rules and the threats to validity in advance, so that the evaluation cannot be shaped by the results.

5.8.1 Evidence Available at FYP-I

The audit (Section 5.5.3), the multi-year retrieval (Section 5.5.4) and the baseline run (Section 5.6.7) support these statements about the locust module, and no more:

- Real point observations exist for the Pakistani part of the study box in early 2020 (887 records), so the evaluation-period signal is real, and the label-building code was developed on actual data.
- The records are heavily clustered (21 reports in the 2020 extracts), concentrated in spring, and dominated by one hopper surge in April, so the evaluation period contains a small number of independent events and any conclusion will have wide uncertainty.
- Survey-confirmed absences exist in every year from 2007 to 2026, so survey-confirmed negatives are a realistic option.
- The multi-year archive was retrieved. Positives are rare before the 2019–2020 upsurge, and under rolling-origin evaluation the environment-only baselines do not beat chance while a rule based on recent sightings does.

No claim about the skill of the ConvLSTM, or about anomaly-ranking quality or classification accuracy of the crop module, is made.

5.8.2 Experimental Plan for FYP-II

Table 5.16: Planned experiments

ID	Question	Setup	Key metric
E1	How well do simple models forecast presence? (done for B0–B2b and persistence; B3 pending)	Baselines on rolling-origin forecasts over 2019–2020 and on the frozen splits, all three horizons	PR-AUC, F1, recall at alert area
E2	Does the ConvLSTM beat the baselines? (RQ1)	ConvLSTM vs. persistence, B2, B3 under rolling-origin forecasts; five seeds	PR-AUC with block-bootstrap interval
E3	How much do negative examples matter? (RQ2)	N1, N2, N3 with the same model and protocol; N1 only so far	PR-AUC change; calibration
E4	Does the target definition matter? (done for baselines)	Any-stage target (primary) vs. breeding-stage target (secondary)	PR-AUC per horizon
E5	Which inputs matter? (past-record input done for tabular models)	Remove channels; add mask channels; add past-record input (labelled separately)	PR-AUC change

ID	Question	Setup	Key metric
E6	Does the Isolation Forest rank wheat fields better than simple rules? (RQ3)	IF vs. peer-median deviation vs. NDVI-only rule	PR-AUC / Precision@k
E7	What drives the crop score?	Global vs. per-group forests; NDVI-only vs. multiband features; 200/500/1000 trees	Change in ranking quality; stability across seeds
E8	Can the crop module warn early, and is it fooled by late sowing?	As-of-date scoring; flood-delayed-district check	Date at which scores separate; score shift
E9	How much local evidence is enough? (RQ4)	Supervised random forest if $\geq$200 labelled fields; foreign model zero-shot	F1, PR-AUC vs. number of labels

5.8.3 Reporting Rules

Results will be reported with mean and standard deviation over five seeds and with block-bootstrap intervals; the test set is evaluated once after all choices are frozen; negative and null results are reported with the same prominence as positive ones; and every claim is labelled with its evidence tier (Section 5.7.1). The success criterion for the ConvLSTM (Section 5.6.6) was fixed before any ConvLSTM result existed for this reason; its one change, adding persistence as a comparator, is stated there.

5.8.4 Threats to Validity

- **Construct validity.** FAO records measure where surveyors found locusts, not where locusts bred; "abnormal" crop development is not disease. Both limits are stated wherever results are given.
- **Internal validity.** Leakage through splits, climatology, peer statistics or threshold selection would inflate results; the checks in Table 6.1 address this, and all learned quantities are fitted without evaluation-period data.
- **External validity.** The locust evaluation is one real event in one region, and the public wheat survey covers a single season, so generalisation across years, districts or crops is not established by this project; the third evidence tier will probably remain unsupported.
- **Statistical validity.** With few independent test events and clustered records, intervals are wide; differences smaller than the interval are not claimed.

5.9 Work Plan, Status and Lessons

This section gives the status of the objectives, what was learned so far, the limitations of the current design and the schedule for the next phase.

Table 5.17: Status of the objectives at the end of FYP-I

ID	Objective (short)	Status
O1	Locust dataset (box covering Cholistan, Tharparkar and part of Rajasthan)	Built: labels from 2008, inputs from 2001; documentation for release pending
O2	Baselines and ConvLSTM	B0–B2b and the persistence and sightings baselines implemented and evaluated; B3 and the ConvLSTM specified, not yet implemented
O3	Rolling-origin 2019–2020 evaluation	Protocol implemented and run for the baselines; ConvLSTM not yet run
O4	Wheat cohort from public data	Rules defined; source located; extraction pending
O5	Features and Isolation Forest	Fully specified; not yet implemented
O6	Local calibration or Precision@k	Protocol defined; data request pending
O7	Dashboard and lead-time comparison	Wireframe designed; implementation pending

5.9.1 Status of the Objectives

5.9.2 Challenges and Lessons

- **Finding the data.** The obvious portal returned no data layers; the open catalogue and its query parameters were found only by inspecting what each download actually contained. Data access is a project task in its own right, not a preliminary; the decisive step was reading the catalogue's query page, which showed that the endpoint accepts SQL over the whole master view.
- **Data-quality traps.** A mislabelled country column, a timestamp that looks like a date, an unreliable breeding flag and heavily clustered records would each have silently corrupted a naive pipeline; auditing real files before designing models paid off.
- **Design changes.** The design changed four times in FYP-I (Table 5.3), each for a concrete reason; keeping the reasons written down made the changes defensible.
- **Labels are the bottleneck.** Locust labels are presence-only and scarce outside the 2019–2020 upsurge; crop-health labels are not public. Both shape what can be claimed.
- **Honest baselines first.** Environment-only baselines do not beat chance, while a parameter-free persistence rule does. Without the baseline ladder and the rolling-origin protocol, a ConvLSTM score would have been hard to interpret.

5.9.3 Limitations

The report presents results only for the locust baselines; the ConvLSTM, the per-cell LSTM and the crop module are untrained. The locust training years contain few positives, 2019–2020 is a single upsurge

with strongly clustered records, the rolling-origin period overlaps the validation and test blocks so that no untouched test set remains, the climatology includes 2019 input data, the sightings baselines assume records are available at forecast time, and the study box covers part of India as well as Pakistan. A parameter-free persistence rule was also scored on the test block during diagnosis (PR-AUC 0.21 against a prevalence of 0.12 at 28 days); no model or setting was chosen on it. The crop module's binary claims depend on local labels that may not be obtained, in which case it is limited to anomaly ranking; its public wheat survey covers one season and its likely validation source covers a different region and years. Field results do not identify causes. The dashboard is a wireframe.

5.9.4 Plan for FYP-II

FYP-II is planned as 16 weeks (Figure 5.8). Weeks 1–2 settle data access, because every later step depends on it.

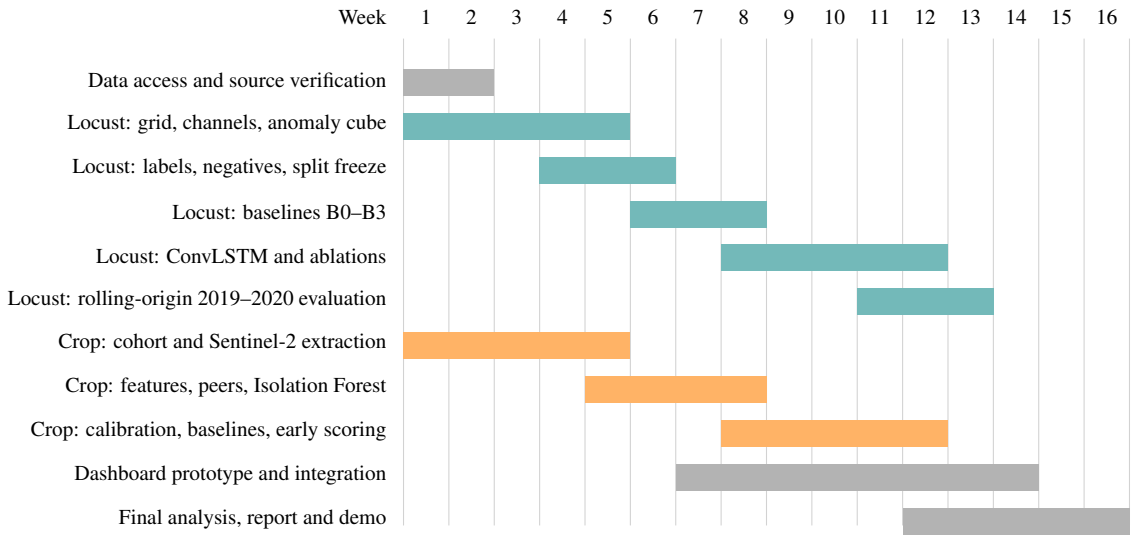


Figure 5.8: FYP-II schedule (weeks): teal = locust module, orange = crop module, grey = shared

Immediate next steps. (1) Agree the locust design changes of Table 5.3 (any-stage primary target, training targets 2008–2019, rolling-origin evaluation, ERA5-Land inputs, the sightings baseline) with the supervisor; (2) implement the per-cell LSTM and train the ConvLSTM under the same rolling-origin protocol with five seeds; (3) run the N2 and N3 negative-example ablations; (4) send the data request for the wheat surveillance records; (5) download and profile the ADB survey and start the Sentinel-2 extraction.

5.9.5 Recommendations for Future Work

If time and data allow: fine-tune a geospatial foundation model for the locust task [10] as a comparison; extend the locust grid to neighbouring countries in the recession area; evaluate a contrastive model

Table 5.18: FYP-II milestones

ID	Week	Milestone	Status
M1	2	Data access settled: multi-year FAO archive retrieved (or fallback chosen); positives counted; ADB survey profiled; product coverage verified; surveillance-data request sent	Locust part done; ADB profiling and surveillance request pending
M2	6	Datasets frozen: locust cube, labels and split; crop cohort manifest and features	Locust part done; crop pending
M3	8	Baselines B0–B3 and crop baselines evaluated	Locust B0–B2b and persistence done; B3 and crop pending
M4	12	ConvLSTM, Isolation Forest, ablations and calibration complete	Pending
M5	13	Rolling-origin 2019–2020 evaluation complete	Baselines done; models pending
M6	14	Dashboard integrated with both modules	Pending
M7	16	Final report and demonstration	Pending

for the crop module [28]; add Sentinel-1 radar to the crop features [4]; and, with agronomist partners, collect verified field-health labels so that the crop module can move from anomaly ranking to calibrated screening.

5.10 Conclusion

This chapter specified a two-module framework with a common principle and distinct models. The locust module is a supervised forecaster trained on a masked-label dataset we construct, evaluated on a real invasion by rolling-origin forecasts and embargoed time blocks; the crop module is an unsupervised, peer-normalised Isolation Forest whose Normal/Abnormal output is claimed only after local calibration. Design changes made during FYP-I, the evidence tiers and the explicit success criteria keep the claims matched to the evidence. Chapter 6 describes how these pipelines are organised as a system.

Chapter 6 High-Level and Low-Level Design

This chapter describes how the methodology of Chapter 5 is organised as a system: the layers and subsystems, the design considerations and strategies behind them, the main classes of the prototype, and the planned dashboard. AgriWatch is a research prototype, so the design emphasises reproducibility and traceable evidence over deployment concerns such as user accounts or scalability.

6.1 System Overview

AgriWatch is a batch data-and-modelling pipeline with a thin presentation layer. Raw data from open sources are acquired and frozen; two independent processing pipelines turn them into model-ready arrays (locust) and a field feature table (crop); models and baselines are trained and scored by a shared evaluation harness that enforces the splits; and the resulting risk maps and field scores are shown in a dashboard together with the level of evidence behind them. Nothing runs in real time. Figure 6.1 shows the layered structure.

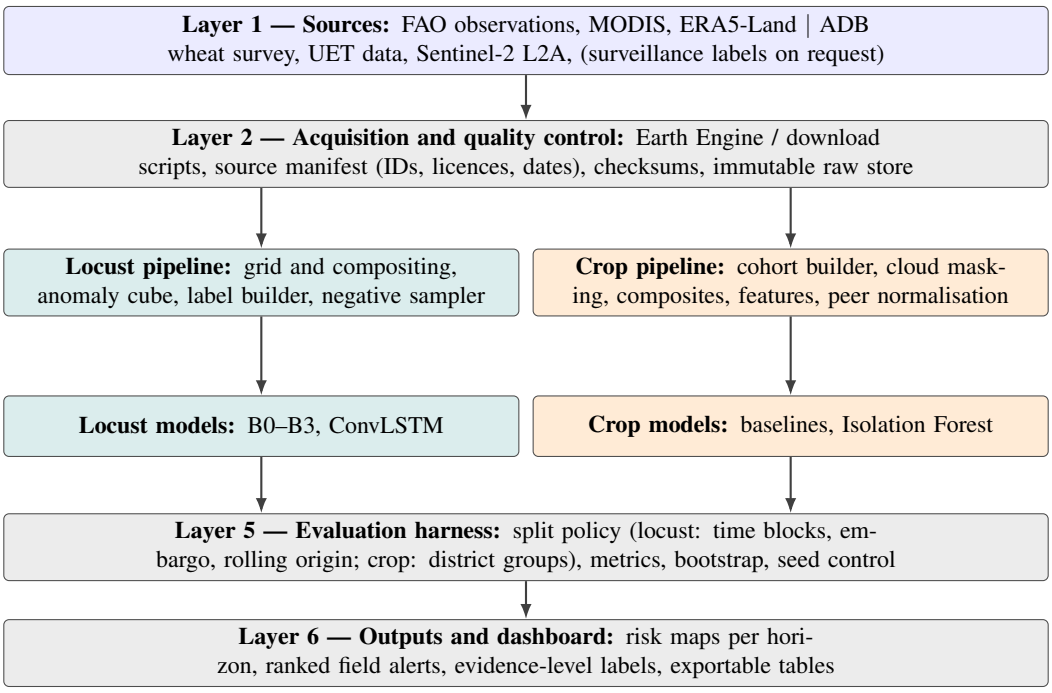


Figure 6.1: Layered architecture of the AgriWatch prototype

6.2 Design Considerations

6.2.1 Assumptions and Dependencies

- The FAO observation data remain available through the open catalogue (CC BY 4.0), which supplied the multi-year records [44]; the Locust Hub request [39] is no longer needed.
- Earth Engine access for research continues to be free, and Sentinel-2 L2A covers the wheat seasons used (Rabi 2022–23 is well within the L2A archive) [45, 9].
- The ADB survey record stays downloadable [33], and verified wheat-health observations can be obtained by request [3]; if they cannot, the crop module's claims fall back to tier 1 (Section 5.7.1).
- Model training uses a laptop and free-tier cloud GPUs; Python is the implementation language, with scikit-learn [46] for classical models and a deep-learning framework for the ConvLSTM.
- The users are analysts and researchers, not field operators; outputs are interpreted as risk or inspection priorities by people with domain knowledge.

6.2.2 General Constraints

- **Environment and resources:** no GPU cluster, no field equipment; exports must fit Earth Engine quotas.
- **Licensing and compliance:** FAO data are used for non-commercial research with acknowledgement; each source's licence is recorded in the manifest.
- **Privacy and ethics:** survey points are locations of private farms. The dashboard and any released tables show field-level results only in aggregated or anonymised form (for example district-level summaries or rounded coordinates), and no result is presented as a diagnosis of a particular farmer's field.
- **Reproducibility:** every table and figure must be regenerable from versioned scripts, a configuration file and the source manifest.
- **Verification:** automated checks (Section 6.6.2) must show that splits are leak-free before any score is reported.

6.2.3 Goals and Guidelines

- **Simplest model that answers the question.** Each model is justified against the simpler one before it (the baseline ladder).
- **Evidence before claims.** Every output carries the evidence level behind it (Section 5.7.1), and negative results are reported.
- **Interpretability.** Features are named indices and temporal summaries rather than opaque embeddings, so that an alert can be explained.
- **Modularity.** Locust and crop pipelines share only the index library and the evaluation harness, so each can progress independently.

6.2.4 Development Methods

Development is iterative with a "crawl before run" discipline: a data loader and sanity check come first, then the simplest baseline, and only then the more complex model. Each iteration ends with a checkpoint on which results and decisions are recorded. Source code and configuration are kept under version control; exploratory notebooks are converted to scripts once a step is stable; random seeds, scene identifiers and software versions are logged. A heavier, formal process was not adopted because the work is exploratory and the team is small.

6.3 System Architecture

The architecture of Figure 6.1 decomposes the system by *pipeline* (locust, crop) rather than by technical role (all data handling together, all models together). The reason is that the two problems differ in data, labels and models, so a role-based split would couple them without benefit; a pipeline-based split lets either module fail, change or be dropped without affecting the other. The components are:

- **Acquisition and manifest:** scripts that pull data and write a manifest of source, licence, date range, coordinate system and checksum; raw data are never edited in place.
- **Processing pipelines:** deterministic transforms from raw data to analysis-ready arrays (locust) or feature tables (crop), configured from one file.
- **Model zoo:** baselines and main models behind a common interface (fit, predict, save).
- **Evaluation harness:** the only component that creates splits and computes metrics, so that every model is evaluated identically.

- **Artefact store and dashboard:** risk maps, scores and tables, and the interface that displays them.

6.3.1 Subsystem Architecture

Locust subsystem. (1) *GridSpec* defines the bounding box, cell size and time step. (2) Channel extractors write a NetCDF-style array of shape (time, channel, rows, columns). (3) The anomaly stage fits climatology on 2001–2019 and writes the anomaly cube plus a mask channel. (4) The label builder implements Algorithm 1 and writes Y_h and M_h for each origin. (5) Models read the cube and labels, and the harness evaluates them on the frozen splits.

Crop subsystem. (1) The cohort builder reads the ADB survey, selects wheat records and applies the inclusion rules, writing a manifest with exclusion reasons. (2) The Sentinel-2 extractor produces composites and zonal statistics per field–season. (3) The feature builder derives temporal summaries and peer groups. (4) The scorer fits peer statistics and the Isolation Forest on the training partition and scores all fields. (5) The calibration module chooses a threshold only when labelled validation fields exist.

Dashboard subsystem. A read-only interface over the artefact store with three screens—overview map (S1), wheat field alerts (S2) and evidence and data (S3). Their wireframes, navigation and mapping to use cases are specified in Chapter 4. It is a prototype to demonstrate the combined outputs, not a public service.

6.4 Architectural Strategies

6.4.1 Strategy 1: Server-Side Reduction in Earth Engine

Decision. Reduce 20 years of imagery to small per-cell and per-field arrays inside Earth Engine and export only those. *Alternatives.* Downloading raw tiles to a laptop was rejected: the volume is prohibitive. Local processing of full-resolution mosaics was rejected for the same reason. *Consequence.* Exports are asynchronous and quota-limited, so the pipeline caches intermediate arrays and treats export failures as recoverable.

6.4.2 Strategy 2: Manifest-Driven Reproducibility

Decision. Every dataset and derived table is described in a manifest (source, licence, access date, coordinate system, checksum, version), and results are generated from versioned configuration. *Alternatives.* Ad hoc notebooks were rejected because they make it impossible to reproduce a number or to prove that a split was leak-free. *Consequence.* A small overhead per data source, repaid when results must be

regenerated.

6.4.3 Strategy 3: Two Pipelines Sharing Only an Index Library and the Harness

Decision. Keep the locust and crop pipelines independent and share only the spectral-index functions and the evaluation harness. *Alternatives.* A single "shared anomaly engine" was considered earlier; it was dropped because the modules compute different things—locust anomalies are temporal departures from a cell's own history, crop anomalies are cross-sectional departures from peers—so a shared engine would be mostly adapter code. *Consequence.* Duplication is small, and either module can be developed without waiting for the other.

6.4.4 Strategy 4: Separation of Unlabelled and Labelled Data

Decision. Treat Dataset A (unlabelled cohort) and Dataset B (verified health labels) as separate artefacts with separate access conditions. *Alternatives.* Mixing foreign labelled data into the training set was rejected because regional shift makes it unreliable [12]. *Consequence.* Without Dataset B the module still runs and still produces scores, but its claims are limited to anomaly ranking by design.

6.5 Domain Model / Class Diagram

Figure 6.2 shows the main classes of the prototype and their relations. *DataSource* records provenance and licence for every input. In the locust chain, *FaoObservationStore* holds the cleaned records, *GridSpec* defines the grid, *AnomalyCube* holds the channel arrays, and *LabelBuilder* produces masked labels; the abstract *ForecastModel* is implemented by the baselines B0–B3, the recent-sightings baselines and the ConvLSTM. In the crop chain, *WheatCohort* lists field–season units, *S2Extractor* produces composites and zonal statistics, *FeatureTable* adds temporal features and peer groups, and *AnomalyScorer* wraps the Isolation Forest and the threshold calibration. *SplitPolicy* and *Evaluator* are shared, and *Dashboard* reads their outputs.

6.6 Policies and Tactics

6.6.1 Coding, Configuration and Versioning

Code follows common Python conventions (PEP 8 naming, docstrings for public functions), is organised as importable modules with thin command-line scripts, and is kept under version control. Parameters (study box, cell size, horizons, split dates, model settings) live in one configuration file that is copied

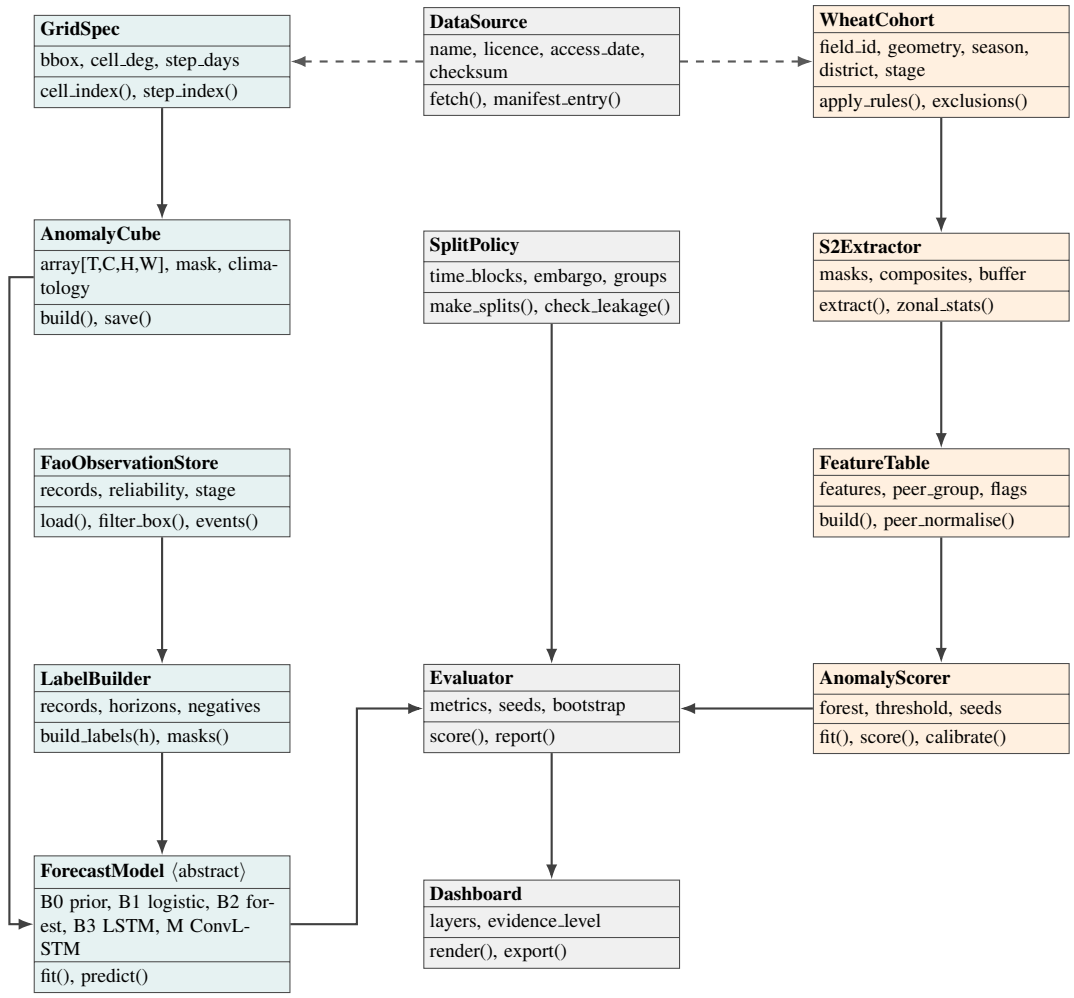


Figure 6.2: Main classes and relations of the prototype (solid: data flow; dashed: provenance)

into every results folder. *Alternatives considered:* parameters hard-coded in notebooks (rejected: unreproducible).

6.6.2 Testing Tactics

Tests target the places where errors would silently inflate results: label windows must not overlap inputs; training targets must not reach the validation period; climatology must use training years only; every rolling-origin refit must use only labels whose target windows had ended; in the crop module no district may appear in more than one partition; and a data-selection regression test must reproduce the known audit counts (Table 6.1). Statistical results are never accepted until these pass.

Table 6.1: Verification checks

ID	Component	Check	Expected	Status
T1	Label builder	Latest input date is earlier than the start of every target window	No overlap	Executed
T2	Split policy (locust)	No training origin has a target window ending after the validation start (embargo, per horizon), and likewise for validation/test	Zero violations	Executed

ID	Component	Check	Expected	Status
T3	Anomaly cube	Climatology uses only training years	Years used = 2001–2019	Executed
T4	Grid snapping	Records on cell borders are assigned deterministically and consistently	Same cell on re-run	Executed
T5	Record selection	In the 2020 extracts, rows with FAO code `PA` inside the study box are counted (audit only; the pipeline selects by coordinates)	887 records, 729 distinct coordinates	Executed in audit
T6	Index functions	NDVI, NDRE, GNDVI, NDMI and OSAVI on synthetic reflectances with known answers	Match hand calculation	Planned
T7	Split policy (crop)	No district appears in more than one partition	Zero violations	Planned
T8	Isolation Forest	Same data and seed give identical scores	Identical scores	Planned
T9	Masked loss	Unknown cells ($m = 0$) contribute zero gradient	Zero change	Planned
T10	Baselines	Probabilities lie in $[0, 1]$ and are finite for all known cells	No invalid values	Executed
T11	Rolling origin	Every rolling-refit training sample has a target window that ended before the origin it is used for, and no evaluation sample is in the training set of its own group	Zero violations	Executed

6.6.3 Data Handling and Compliance

Raw files are read-only and checksummed; derived tables are regenerated, not edited; licences and acknowledgement text are stored with each source. Any public release of crop-field results is aggregated or anonymised.

6.7 Conclusion

The design organises two independent pipelines behind a shared evaluation harness and a read-only dashboard, with provenance recorded for every input and leakage checks enforced by tests. Taken together, Chapters 1–6 define AgriWatch as a single R&D project with two modules that share one principle and one evaluation discipline: the Locust Risk Module (a supervised ConvLSTM forecaster trained on a dataset constructed from FAO records) and the Crop Abnormality Screening Module (an unsupervised, peer-normalised Isolation Forest on Sentinel-2 wheat time series, calibrated only with local evidence). What remains is to execute the work plan of Section 5.9: train the ConvLSTM and the remaining locust models, build the crop cohort and features, run the remaining experiments of Section 5.8, and implement the dashboard specified in Chapter 4.

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Chapter A FAO Field Dictionary

Table A.1 lists the FAO fields of the 2020 extracts used or deliberately not used in FYP-I. The multi-year retrieval (Section 5.5.4) used the master view, whose few columns (category, latitude, longitude, start date, country name, location name, area treated, control) are a subset of these. The life-stage files share one 152-column schema; the remaining columns describe hopper and adult group characteristics and control operations and are largely empty or zero in the Pakistani study-box records.

Table A.1: FAO fields: use in the project

Field	Meaning (as observed)	Use
`Lon`, `Lat`	Record coordinates (WGS84)	Used; grid snapping and study-box selection (never country selection)
`Observation Date`	Date of the observation	Used; label timing
`Location Reliability`	Reliability of the location ("Exact" for all in-box records)	Used as a filter
`Location Name`	Free-text place name (some are only a province name)	Informational; not used for location
`Impacted Area (ha)`	Area occupied (median 200 ha, maximum 5,000 ha in-box)	Informational
`Report ID`	Identifier of the survey report	Used to assess clustering
`PARTMONTH`	Early / middle / late part of month	Informational
`Country ISO2 Code`	FAO legacy country code (Pakistan = `PA`)	*Not* used for selection; used only in the audit to count Pakistani rows
`country_name` (master view)	Country name in the catalogue's master view	*Not* used for selection; used only to report counts by country
`LOCPRESENT`	1 for all but five rows; the five with value 2 are presumably recorded absences	Records with value 2 excluded
`BREEDING`	1 for five in-box adult records, otherwise 2	Not used; unreliable as a label
`TmSTARTDAT`	Holds the download timestamp, not an observation date	Not used
Remaining fields	Group density, maturity, colour, appearance and control-operation descriptors	Not used in FYP-I

Chapter B Configuration Summary

The experiment configuration that FYP-II will version and copy into every results folder (initial values; changed only on validation evidence):

```
locust:
  box: {lat: [24, 31], lon: [68, 74]}
  cell_deg: 0.25            # 28 rows x 24 columns
  step_days: 16
  input_steps: 12
  channels: [ndvi, water_index, soil_moisture, temperature, rainfall]  # +
      observed-mask channel
  region_selection: coordinates   # never country code or name
  horizons_days: [28, 42, 56]
  target_window_days: 16
  target: any_stage          # adult, hopper, band, swarm; "breeding_stage" (
      hopper, band) is secondary
  input_years: [2001, 2020]
  climatology_years: [2001, 2019]
  anomaly_clip: 5.0
  mask_min_valid_fraction: 0.2
  splits_by_target_window: {train: [2008-01-01, 2019-12-31], val: [2020-01-01,
       2020-06-30],
                            test: [2020-07-01, 2020-12-31]}
  embargo: per_horizon       # straddling target windows dropped (at most 72
      days)
  rolling_origin: {eval_targets: [2019-01-01, 2020-12-31], refit_every_origins
      : 6}
  negatives: n1_survey_absence | n2_stratified | n3_random
  convlstm: {layers: [32, 64], kernel: 3, dropout: 0.1, lr: 1.0e-3,
             weight_decay: 1.0e-4, batch: 8, epochs: 100, patience: 10}
crop:
  season_composite_days: 10
  min_valid_coverage: 0.60
  max_interp_gap_composites: 2
  indices: [ndvi, ndre, gndvi, ndmi, osavi]
  isolation_forest: {trees: 500, max_samples: 256, contamination: auto}
  corr_filter: 0.95
  seeds: [42, 52, 62, 72, 82]
  split: {train: 0.70, val: 0.15, test: 0.15, group_by: district}   # never by
       season
  features: {variables: 15, summaries_per_variable: 8, total_before_filtering:
       120}
```

Chapter C Data Audit Script

The script below produced Table 5.5, Table 5.6, and Figures 5.2 and 5.3 from the downloaded FAO files, and writes every audit number to a JSON file.

```
"""
data_audit.py -- first-pass audit of the FAO Desert Locust observation files.

Reads the four life-stage extracts (adults/hoppers/bands/swarms, 2020 snapshot)
and the rolling 3-month "Global" download, and produces:
  * Figures/fao_study_box_map.pdf   : records over the study box and 0.25-degree grid
  * Figures/fao_monthly_counts.pdf  : monthly record counts per life stage in the box
  * audit_stats.json                : every number quoted in the report

Run:  python data_audit.py <folder-with-csvs>
"""
import json, sys, os
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.patches import Rectangle

SRC = sys.argv[1] if len(sys.argv) > 1 else "."
OUT = os.path.dirname(os.path.abspath(__file__))

# Study box fixed in the project plan (Cholistan + Tharparkar desert margin)
LAT0, LAT1, LON0, LON1 = 24.0, 31.0, 68.0, 74.0
CELL = 0.25
STAGES = ["adults", "hoppers", "bands", "swarms"]
COL = {"adults": "#E69F00", "hoppers": "#0072B2", "bands": "#009E73", "swarms": "#D55E00"}

plt.rcParams.update({"font.family": "STIXGeneral", "mathtext.fontset": "stix",
                     "font.size": 9, "axes.linewidth": 0.6})

def in_box(d):
    return d["Lat"].between(LAT0, LAT1) & d["Lon"].between(LON0, LON1)

frames, stats = {}, {}
for s in STAGES:
    d = pd.read_csv(os.path.join(SRC, f"{s}_2020.csv"), low_memory=False)
    ncols = int(d.shape[1])                       # columns in the file as delivered by FAO
    d = d.assign(stage=s, date=pd.to_datetime(d["Observation Date"], errors="coerce"))
```

```
    frames[s] = d
    pk = d[d["Country ISO2 Code"] == "PA"]          # FAO legacy code: PA = Pakistan
    stats[s] = {
        "rows_all_countries": int(len(d)),
        "columns": ncols,
        "date_min": str(d["date"].min().date()), "date_max": str(d["date"].max().
        ↪ date()),
        "pakistan_rows": int(len(pk)),
        "pakistan_in_box": int(in_box(pk).sum()),
        "locpresent_absent_rows": int((d["LOCPRESENT"] == 2).sum()),
        "pk_iso_rows": int((d["Country ISO2 Code"] == "PK").sum()),
    }

allf = pd.concat(frames.values(), ignore_index=True)
pk = allf[allf["Country ISO2 Code"] == "PA"].copy()
box = pk[in_box(pk)].copy()
stats["pakistan_lat_range"] = [float(pk["Lat"].min()), float(pk["Lat"].max())]
stats["pakistan_lon_range"] = [float(pk["Lon"].min()), float(pk["Lon"].max())]
stats["box_total"] = int(len(box))
stats["box_unique_coords"] = int(box[["Lat", "Lon"]].drop_duplicates().shape[0])
stats["box_reliability"] = {str(k): int(v) for k, v in box["Location Reliability"].
↪ value_counts().items()}
stats["box_report_ids"] = int(box["Report ID"].nunique())
named_prov = box["Location Name"].isin(["Punjab", "Sindh", "Balochistan"])
stats["box_province_label_rows"] = int(named_prov.sum())
stats["box_punjab_label_rows"] = int((box["Location Name"] == "Punjab").sum())
stats["box_punjab_label_unique_coords"] = int(
    box[box["Location Name"] == "Punjab"][["Lat", "Lon"]].drop_duplicates().shape
    ↪ [0])

# monthly counts inside the box
box["month"] = box["date"].dt.to_period("M").astype(str)
mt = box.groupby(["month", "stage"]).size().unstack(fill_value=0).reindex(columns=
↪ STAGES, fill_value=0)
stats["box_monthly"] = {m: {s: int(mt.loc[m, s]) for s in STAGES} for m in mt.index}

# rolling-window "Global" download
r = pd.read_csv(os.path.join(SRC, "result.csv"))
r["start_date"] = pd.to_datetime(r["start_date"])
rp = r[r["country_name"] == "Pakistan"]
stats["rolling"] = {
    "rows": int(len(r)), "date_min": str(r["start_date"].min().date()),
    "date_max": str(r["start_date"].max().date()),
    "categories": {k: int(v) for k, v in r["category"].value_counts().items()},
    "pakistan_rows": int(len(rp)),
    "pakistan_categories": {k: int(v) for k, v in rp["category"].value_counts().
    ↪ items()},
    "pakistan_in_box": int((rp["lat"].between(LAT0, LAT1) & rp["lon"].between(LON0,
    ↪ LON1)).sum()),
```

```
}
with open(os.path.join(OUT, "audit_stats.json"), "w") as f:
    json.dump(stats, f, indent=2)

# ---------------- Figure 1: map of records over study box + grid ----------------
fig, ax = plt.subplots(figsize=(6.2, 4.9))
for lon in np.arange(LON0, LON1 + 1e-9, CELL):
    ax.plot([lon, lon], [LAT0, LAT1], color="#d9d9d9", lw=0.25, zorder=0)
for lat in np.arange(LAT0, LAT1 + 1e-9, CELL):
    ax.plot([LON0, LON1], [lat, lat], color="#d9d9d9", lw=0.25, zorder=0)
ax.add_patch(Rectangle((LON0, LAT0), LON1 - LON0, LAT1 - LAT0, fill=False, ec="black
↪ ", lw=1.2, zorder=3))
outside = pk[~in_box(pk)]
ax.scatter(outside["Lon"], outside["Lat"], s=5, c="#9e9e9e", alpha=0.55, lw=0,
↪ zorder=2,
           label="Pakistan, outside study box")
for s in STAGES:
    d = box[box["stage"] == s]
    if len(d):
        ax.scatter(d["Lon"], d["Lat"], s=11, c=COL[s], alpha=0.85, lw=0, zorder=4,
                   label=f"{s.capitalize()} (n={len(d)})")
ax.set_xlim(61, 75); ax.set_ylim(23.2, 33.2)
ax.set_xlabel("Longitude (degrees E)"); ax.set_ylabel("Latitude (degrees N)")
ax.set_aspect("equal")
ax.legend(loc="upper left", fontsize=7.5, frameon=True, framealpha=0.95, markerscale
↪ =1.4)
fig.tight_layout(); fig.savefig(os.path.join(OUT, "Figures", "fao_study_box_map.pdf"
↪ )); plt.close(fig)

# ---------------- Figure 2: monthly counts per stage in the box ----------------
months = list(mt.index)
labels = [pd.Period(m).strftime("%b %Y") for m in months]
x = np.arange(len(months)); w = 0.2
fig, ax = plt.subplots(figsize=(6.2, 3.1))
for i, s in enumerate(STAGES):
    ax.bar(x + (i - 1.5) * w, mt[s].values, width=w, color=COL[s], label=s.
    ↪ capitalize(), lw=0)
ax.set_xticks(x); ax.set_xticklabels(labels)
ax.set_ylabel("Records inside study box")
ax.set_ylim(0, 420)
ax.legend(fontsize=7.5, ncol=4, loc="upper left", frameon=False)
ax.spines[["top", "right"]].set_visible(False)
fig.tight_layout(); fig.savefig(os.path.join(OUT, "Figures", "fao_monthly_counts.pdf
↪ ")); plt.close(fig)

print(json.dumps({k: stats[k] for k in ["box_total", "box_unique_coords", "
↪ box_monthly", "rolling"]}, indent=1))
```